
LOCAL CLASS FIELD THEORY

par

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Introduction

We call a field K a local field if it is complete with respect to the topology defined by a discrete valuation v and if its residue field k is finite. We write $q = p^f = \text{Card}(k)$ and we always assume that the valuation v is normalized; that is, that the homomorphism $v : K^\times \rightarrow \mathbf{Z}$ is surjective. The structure of such fields is known:

1. If K has characteristic 0, then K is a finite extension of the p -adic field $\mathbf{Q}_p$, the completion of $\mathbf{Q}$ with respect to the topology defined by the p -adic valuation. If $[K : \mathbf{Q}_p] = n$, then $n = ef$, where f is the residue degree, that is, $f = [k : \mathbf{F}_p]$, and e is the ramification index $v(p)$.
2. If K has characteristic p ("the equal characteristic case"), then K is isomorphic to a field $k((T))$ of formal power series, where T is a uniformizing parameter.

The first case is the one which arises in completions of a number field relative to a prime number p .

We shall study the Galois groups of extensions of K and would of course like to know the structure of the Galois group $G(K_s/K)$ of the separable closure K_s of K , since this contains the information about all such extensions.

(In the case of characteristic 0, $K_s = \overline{K}$.) We shall content ourselves with the following:

1. The cohomological properties of all Galois extensions, whether abelian or not.
2. The determination of the abelian extensions of K , that is, the determination of G modulo its derived group G' .

Throughout this chapter, we shall adhere to the notation already introduced above, together with the following. We denote the ring of integers of K by $\mathcal{O}_K$, the multiplicative group of K by $K^\times$, and the group of units by U_K . A similar notation will be used for extensions L of K , and if L is a Galois extension, then we denote the Galois group by $G(L/K)$, or $G_{L/K}$, or even by G . If $s \in G$ and $x \in L$, then we denote the action of s on x by ${}^s x$ or by $s(x)$.

In addition to the preceding Chapters, the reader is referred to "Corps Locaux" (Actualités scientifiques et industrielles, 1296; Hermann, Paris, 1962) for some elided details. In what follows theorems etc. in the four sections are numbered independently,

1. The Brauer Group of a Local Field

1.1. Statements of Theorems. — In this first section, we shall state the main results; the proofs of the theorems will extend over §§ 1.2–1.6.

We begin by recalling the definition of the Brauer group, $\text{Br}(K)$, of K .

(See Chapter V, § 2.8.) Let L be a finite Galois extension of K with Galois group $G(L/K)$. We write $H^2(L/K)$ instead of $H^2(G(L/K), L^\times)$ and we consider the family $(L_i)_{i \in I}$ of all such finite Galois extensions of K . The inductive (direct) limit

$$\varinjlim_i H^2(L_i/K)$$

is by definition the Brauer group $\text{Br}(K)$ of K .

It follows from the definition that $\text{Br}(K) = H^2(K_s/K)$. In order to compute $\text{Br}(K)$ we look first at the intermediate field

$$K \subset K_{\text{nr}} \subset K_s,$$

where K_{nr} denotes the maximal unramified extension of K . The reader is referred to Chapter I, § 7 for the properties of K_{nr} . We recall in particular that the residue field of K_{nr} is $\bar{k}$, the algebraic closure of k , and that

$$G(K_{\text{nr}}/K) = G(\bar{k}/k).$$

We denote by F the Frobenius element in $G(K_{\text{nr}}/K)$; the effect of F on the residue field $\bar{k}$ is given by $x \mapsto x^q$. The map $n \mapsto F^n$ is an isomorphism

$$\widehat{\mathbf{Z}} \xrightarrow{\sim} G(K_{\text{nr}}/K)$$

of topological groups. From Chapter V

§ 1.5, we recall that $\widehat{\mathbf{Z}}$ is the projective (inverse) limit $\varprojlim \mathbf{Z}/n\mathbf{Z}$ of the cyclic groups $\mathbf{Z}/n\mathbf{Z}$.

Since K_{nr} is a subfield of K_s , $H^2(K_{\text{nr}}/K)$ is a subgroup of $\text{Br}(K) = H^2(K_s/K)$.

In fact:

Theorem 1.1. — $H^2(K_{\text{nr}}/K) = \text{Br}(K)$.

We have already noted above that

$$H^2(K_{\text{nr}}/K) = H^2(\widehat{\mathbf{Z}}, K_{\text{nr}}^\times).$$

Theorem 1.2. — *The valuation map $v : K_{\text{nr}}^\times \rightarrow \mathbf{Z}$ defines an isomorphism*

$$H^2(K_{\text{nr}}/K) \xrightarrow{\sim} H^2(\widehat{\mathbf{Z}}, \mathbf{Z}).$$

We have to compute $H^2(\widehat{\mathbf{Z}}, \mathbf{Z})$. More generally let G be a profinite group and consider the exact sequence

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathbf{Q} \longrightarrow \mathbf{Q}/\mathbf{Z} \longrightarrow 0$$

of G -modules with trivial action. The module $\mathbf{Q}$ has trivial cohomology, since it is uniquely divisible, and so the coboundary map yields an isomorphism

$$H^1(G, \mathbf{Q}/\mathbf{Z}) \xrightarrow{\sim} H^2(G, \mathbf{Z}).$$

Now $H^1(G, \mathbf{Q}/\mathbf{Z}) = \text{Hom}(G, \mathbf{Q}/\mathbf{Z})$.

We turn now to $\text{Hom}(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z})$. Let $\phi \in \text{Hom}(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z})$ and define

$$\gamma : \text{Hom}(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z}) \longrightarrow \mathbf{Q}/\mathbf{Z}, \quad \phi \longmapsto \phi(1).$$

It follows from Theorem 1.2 that we have isomorphisms

$$H^2(K_{\text{nr}}/K) \xrightarrow{v} H^2(\widehat{\mathbf{Z}}, \mathbf{Z}) \xleftarrow{\delta} H^1(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z}) = \text{Hom}(\widehat{\mathbf{Z}}, \mathbf{Q}/\mathbf{Z}),$$

where δ is the coboundary map. The map

$$\text{inv}_K : H^2(K_{\text{nr}}/K) \longrightarrow \mathbf{Q}/\mathbf{Z}$$

is now defined by $\text{inv}_K = \gamma \circ \delta^{-1} \circ v$. For future reference, we state our conclusions in the following corollary.

Corollary. — *The map inv_K defines an isomorphism between the groups $H^2(K_{\text{nr}}/K)$ and $\mathbf{Q}/\mathbf{Z}$.*

Since, by Theorem 1.1, $H^2(K_{\text{nr}}/K) = \text{Br}(K)$, we see that we have defined an isomorphism

$$\text{inv}_K : \text{Br}(K) \xrightarrow{\sim} \mathbf{Q}/\mathbf{Z}.$$

If L is a finite extension of K , the corresponding map will be denoted by inv_L .

Theorem 1.3. — *Let L/K be a finite extension of degree n . Then*

$$\text{inv}_L \circ \text{Res}_{L/K} = n \cdot \text{inv}_K.$$

In other words, the following diagram is commutative:

$$\begin{array}{ccc} \text{Br}(K) & \xrightarrow{\text{Res}_{L/K}} & \text{Br}(L) \\ \text{inv}_K \downarrow & & \downarrow \text{inv}_L \\ \mathbf{Q}/\mathbf{Z} & \xrightarrow{\times n} & \mathbf{Q}/\mathbf{Z}. \end{array}$$

For the definition of $\text{Res}_{L/K}$, the reader is referred to Chapter IV § 4 and to Chapter V § 2.8.

Corollary 1. — An element $\alpha \in \text{Br}(K)$ gives 0 in $\text{Br}(L)$ if and only if $n \text{inv}_K(\alpha) = 0$.

Corollary 2. — Let L/K be a Galois extension of degree n . Then $H^2(L/K)$ is cyclic of order n . More precisely, $H^2(L/K)$ is generated by the element $u_{L/K} \in \text{Br}(K)$ whose invariant is $1/n \in \mathbf{Q}/\mathbf{Z}$.

Démonstration. — This follows from the fact that $H^2(L/K)$ is the kernel of $\text{Res}_{L/K}$. $\square$

1.2. Computation of $H^2(K_{\text{nr}}/K)$. — In this section we prove Theorem 1.2. We have to prove that the homomorphism $H^2(K_{\text{nr}}/K) \rightarrow H^2(\widehat{\mathbf{Z}}, \mathbf{Z})$ is an isomorphism.

Proposition 1.4. — Let K_n be an unramified extension of K of degree n and let $G = G(K_n/K)$. Then for all $q \in \mathbf{Z}$ we have:

1. $H^q(G, U_n) = 0$, where $U_n = U_{K_n}$;
2. the map $v : H^q(G, K_n^\times) \rightarrow H^q(G, \mathbf{Z})$ is an isomorphism.

(Theorem 1.2 is evidently a consequence of (2) of Proposition 1.4, since $H^2(K_{\text{nr}}/K) = H^2(\widehat{\mathbf{Z}}, K_{\text{nr}}^\times)$.)

Démonstration. — The fact that (1) implies (2) follows from the cohomology sequence

$$H^q(G, U_n) \longrightarrow H^q(G, K_n^\times) \longrightarrow H^q(G, \mathbf{Z}) \longrightarrow H^{q+1}(G, U_n).$$

It remains to prove (1). Consider the decreasing sequence of open subgroups

$$U_n \supset U_n^1 \supset U_n^2 \supset \dots$$

defined as follows: $x \in U_n^i$ if and only if $v(x - 1) \geq i$. Now let $\pi \in K$ be a uniformizing element; so that $U_n^i = 1 + \pi^i \mathcal{O}_n$, where $\mathcal{O}_n = \mathcal{O}_{K_n}$. Then $U_n = \varprojlim U_n/U_n^i$. The proof will now be built up from the three following lemmas. $\square$

Lemma 1.5. — Let k_n be the residue field of K_n . Then there are galois isomorphisms

$$U_n/U_n^1 \simeq k_n^\times$$

and, for $i \geq 1$,

$$U_n^i/U_n^{i+1} \simeq k_n^+.$$

(By a galois isomorphism we mean an isomorphism which is compatible with the action of the Galois group on either side.)

Démonstration. — Take $\alpha \in U_n$ and map $\alpha \mapsto \bar{\alpha}$, where $\bar{\alpha}$ is the reduction of α into k_n . By definition, $U_n^1 = 1 + \pi \mathcal{O}_n$; so if $\alpha \in U_n^1$ then $\bar{\alpha} = 1$, and the first part of the lemma is proved.

To prove the second part, take $\alpha \in U_n^i$ and write $\alpha = 1 + \pi^i \beta$ where $\beta \in \mathcal{O}_n$. Now map $\alpha \mapsto \bar{\beta}$. We have to show that in this map a product $\alpha\alpha'$ corresponds to the sum $\bar{\beta} + \bar{\beta}'$. By definition,

$$\alpha\alpha' = 1 + \pi^i(\beta + \beta') + \cdots,$$

whence $\alpha\alpha' \mapsto \bar{\beta} + \bar{\beta}'$.

Finally, the isomorphisms are galois since ${}^s\alpha = 1 + \pi^i {}^s\beta$.

□

Lemma 1.6. — *For all integers q and for all integers $i \geq 0$, we have*

$$H^q(G, U_n^i/U_n^{i+1}) = 0.$$

Démonstration. — For $i = 0$, $U_n^0 = U_n$, and the first part of Lemma 1.5 gives

$$H^q(G, U_n/U_n^1) = H^q(G, k_n^\times) = H^q(G_{k_n/k}, k_n^\times).$$

Now for $q = 1$, $H^q(G_{k_n/k}, k_n^\times) = 0$ (“Hilbert Theorem 90”, cf. Chapter V, § 2.6). For $q = 2$, observe that G is cyclic. Since $k_n^\times$ is finite, the Herbrand quotient $h(k_n^\times) = 1$ (cf. Chapter IV, § 8, Prop. 11); hence the result for $q = 2$. For other values of q the result follows by periodicity.

For $i \geq 1$, the lemma follows from Lemma 1.5 and the fact that k_n^+ has trivial cohomology.

The proof of Theorem 1.2 will be complete if we can go from the groups U_n^i/U_n^{i+1} to the group U_n itself and the following lemma enables us to do this.

□

Lemma 1.7. — *Let G be a finite group and let M be a G -module. Let M^i , $i \geq 0$, and $M^0 = M$, be a decreasing sequence of G -submodules and assume that $M = \varprojlim M/M^i$; more precisely, the map from M to the limit is a bijection.*

Then, if, for some $q \in \mathbf{Z}$, $H^q(G, M^i/M^{i+1}) = 0$ for all i , we have $H^q(G, M) = 0$.

Démonstration. — Let f be a q -cocycle with values in M . Since

$$H^q(G, M/M^1) = 0,$$

there exists a $(q-1)$ -cochain ψ_1 of G with values in M such that $f = \delta\psi_1 + f_1$, where f_1 is a q -cocycle in M^1 . Similarly, there exists ψ_2 such that $f_1 = \delta\psi_2 + f_2$, $f_2 \in M^2$, and so on. We construct in this way a sequence (ψ_n, f_n) where ψ_n

is a $(q-1)$ -cochain with values in M^{n-1} and f_n is a q -cocycle with values in M^n , and $f_n = \delta\psi_{n+1} + f_{n+1}$. Set $\psi = \psi_1 + \psi_2 + \dots$.

In view of the hypotheses on M , this series converges and defines a $(q-1)$ -cochain of G with values in M . On summing the equations $f_n = \delta\psi_{n+1} + f_{n+1}$, we obtain $f = \delta\psi$, and this proves the lemma. $\square$

End of proof of Proposition 1.4. — Take M in Lemma 1.7 to be U_n . It follows from Lemma 1.7 and from Lemma 1.6 that the cohomology of U_n is trivial and this completes the proof of Proposition 1.4 and so also of Theorem 1.2. $\square$

1.3. Some Diagrams. —

Proposition 1.8. — *Let L/K be a finite extension of degree n and let L_{nr} (resp. K_{nr}) be the maximal unramified extension of L (resp. K); so that $K_{\text{nr}} \subset L_{\text{nr}}$. Then the following diagram is commutative:*

$$\begin{array}{ccc} H^2(K_{\text{nr}}/K) & \xrightarrow{\text{Res}} & H^2(L_{\text{nr}}/L) \\ \text{inv}_K \downarrow & & \downarrow \text{inv}_L \\ \mathbf{Q}/\mathbf{Z} & \xrightarrow{\times n} & \mathbf{Q}/\mathbf{Z}. \end{array}$$

Démonstration. — Let $\Gamma_K = G(K_{\text{nr}}/K)$ and let F_K be the Frobenius element of Γ_K ; let Γ_L and F_L be defined similarly. We have $F_L = (F_K)^f$ where $f = [l : k]$ is the residue field degree of L/K .

Let e be the ramification index of L/K , and consider the diagram:

$$\begin{array}{ccccccc} H^2(\Gamma_K, K_{\text{nr}}^\times) & \xrightarrow{v_K} & H^2(\Gamma_K, \mathbf{Z}) & \xrightarrow{\delta^{-1}} & \text{Hom}(\Gamma_K, \mathbf{Q}/\mathbf{Z}) & \xrightarrow{\gamma_K} & \mathbf{Q}/\mathbf{Z} \\ \text{Res} \downarrow & & e \text{ Res} \downarrow & & e \text{ Res} \downarrow & & \downarrow \times n \\ H^2(\Gamma_L, L_{\text{nr}}^\times) & \xrightarrow{v_L} & H^2(\Gamma_L, \mathbf{Z}) & \xrightarrow{\delta^{-1}} & \text{Hom}(\Gamma_L, \mathbf{Q}/\mathbf{Z}) & \xrightarrow{\gamma_L} & \mathbf{Q}/\mathbf{Z}, \end{array}$$

where Res is induced by the inclusion $\Gamma_L \rightarrow \Gamma_K$, and γ_K (resp. γ_L) is given by $\phi \mapsto \phi(F_K)$ (resp. $\phi \mapsto \phi(F_L)$). The three squares (1), (2), (3) extracted from that diagram are commutative; for (1), this follows from the fact that v_L is equal to $e \cdot v_K$ on $K_{\text{nr}}^\times$; for (3), it follows from $F_L = F_K^f$ and $n = ef$; for (2), it is obvious.

On the other hand, the definition of $\text{inv}_K : H^2(\Gamma_K, K_{\text{nr}}^\times) \rightarrow \mathbf{Q}/\mathbf{Z}$ is equivalent to

$$\text{inv}_K = \gamma_K \circ \delta^{-1} \circ v_K,$$

and similarly

$$\text{inv}_L = \gamma_L \circ \delta^{-1} \circ v_L.$$

Proposition 1.8 is now clear. $\square$

Corollary 1.9. — *Let $H^2(L/K)_{\text{nr}}$ be the subgroup of $H^2(K_{\text{nr}}/K)$ consisting of those $\alpha \in H^2(K_{\text{nr}}/K)$ which are “killed by L ” (that is, which give 0 in $\text{Br}(L)$). Then $H^2(L/K)_{\text{nr}}$ is cyclic of order n and is generated by the element $u_{L/K}$ in $H^2(K_{\text{nr}}/K)$ such that $\text{inv}_K(u_{L/K}) = 1/n$.*

Démonstration. — Note that if L/K is Galois then a less violent definition of $H^2(L/K)_{\text{nr}}$ is provided by

$$H^2(L/K)_{\text{nr}} = H^2(L/K) \cap H^2(K_{\text{nr}}/K).$$

Consider the exact sequence

$$0 \longrightarrow H^2(L/K)_{\text{nr}} \longrightarrow H^2(K_{\text{nr}}/K) \xrightarrow{\text{Res}} H^2(L_{\text{nr}}/L).$$

The kernel of the map $H^2(K_{\text{nr}}/K) \rightarrow H^2(L_{\text{nr}}/L)$ is $H^2(L/K)_{\text{nr}}$ and this goes to 0 under $\text{inv}_L : H^2(L_{\text{nr}}/L) \rightarrow \mathbf{Q}/\mathbf{Z}$. On the other hand, it follows from Proposition 1.8 that $\text{inv}_L \circ \text{Res} = n \cdot \text{inv}_K$. The kernel of the latter is $(1/n)\mathbf{Z}/\mathbf{Z}$ and so $H^2(L/K)_{\text{nr}}$ is cyclic of order n , and is generated by $u_{L/K} \in H^2(K_{\text{nr}}/K)$ with $\text{inv}_K(u_{L/K}) = 1/n$. $\square$

Corollary 1.10. — *The order of $H^2(L/K)$ is a multiple of n .*

Démonstration. — $H^2(L/K)$ contains a cyclic subgroup of order n by Corollary 1. $\square$

1.4. Construction of a Subgroup with Trivial Cohomology. — Let L/K be a finite galois extension with Galois group G , where L and K are local fields. According to the discussion in Proposition 1.4, the G -module U_L has trivial cohomology when L is unramified.

Proposition 1.11. — *There exists an open subgroup V of U_L with trivial cohomology. That is, $H^q(G, V) = 0$ for all q .*

Démonstration. — We shall give two proofs; the first one works only in characteristic 0, the second works generally.

Method 1. The idea is to compare the multiplicative and the additive groups of L . We know that L^+ is a free module over the algebra $K[G]$. That is, there exists $\alpha \in L$ such that $[^s\alpha]_{s \in G}$ is a basis for L considered as a vector space over K .

Now take the ring $\mathcal{O}_K$ of integers of K and define

$$A = \sum_{s \in G} \mathcal{O}_K {}^s\alpha.$$

This is free over G and so has trivial cohomology. Moreover, by multiplying α by a sufficiently high power of the local uniformizer π_K , we may take such an A to be contained in any given neighbourhood of 0.

It is a consequence of Lie theory that the additive group of L is locally isomorphic to the multiplicative group. More precisely, the power series

$$e^x = 1 + x + \cdots + x^n/n! + \cdots$$

converges for $v(x) > v(p)/(p-1)$. Thus in the neighbourhood $v(x) > v(p)/(p-1)$ of 0, L^+ is locally isomorphic to $L^\times$ under the map $x \mapsto e^x$. (Note that, in the same neighbourhood, the inverse mapping is given by $\log(1+x) = x - x^2/2 + x^3/3 - \cdots$.) Now define $V = e^A$; it is clear that V has trivial cohomology.

The foregoing argument breaks down in characteristic p ; namely at the local isomorphism of L^+ and $L^\times$.

Method 2. We start from an A constructed as above: $A = \sum_{s \in G} \mathcal{O}_K^s \alpha$. We may assume that $A \subset \mathcal{O}_L$. Since A is open in $\mathcal{O}_L$, $\pi_K^N \mathcal{O}_L \subset A$ for a suitable N . Choose $i \geq N+1$ and set $M = \pi_K^i A$. Then $M \cdot M \subset \pi_K M$. For $M \cdot M = \pi_K^{2i} A \cdot A \subset \pi_K^{2i} \mathcal{O}_L$ and if $i \geq N+1$ then

$$\pi_K^{2i} \mathcal{O}_L \subset \pi_K \cdot \pi_K^i A = \pi_K M.$$

Now let $V = 1 + M$. Then V is an open subgroup of U_L . It remains to be proved that V has trivial cohomology. We define a filtration of V by means of subgroups $V^i = 1 + \pi_K^i M$, $i \geq 0$. (Note that V^i is a subgroup since

$$(1 + \pi_K^i x)(1 + \pi_K^i y) = 1 + \pi_K^i(x + y + \pi_K^i xy),$$

etc.) This yields a decreasing filtration

$$V = V^0 \supset V^1 \supset V^2 \supset \cdots.$$

As in § 1.2, Lemma 1.6, we are reduced to proving that $H^q(G, V^i/V^{i+1}) = 0$ for all q . Take $x = 1 + \pi_K^i \beta$, $\beta \in M$, and associate with this its image $\bar{\beta} \in M/\pi_K M$. This is a group isomorphism of V^i/V^{i+1} and $M/\pi_K M$ and we know that the latter has trivial cohomology, since it is free over G .

This completes our proofs of Proposition 1.11. $\square$

We recall the definition of the Herbrand quotient $h(M)$. Namely,

$$h(M) = \frac{\text{Card}(\widehat{H}^0(M))}{\text{Card}(H^1(M))},$$

when both sides are finite. (See Chapter IV, § 8.)

Corollary 1.12. — *Let L/K be a cyclic extension of degree n . Then we have $h(U_L) = 1$ and $h(L^\times) = n$.*

Démonstration. — Let V be an open subgroup of U_L with trivial cohomology (cf. Proposition 1.11). Since h is multiplicative, $h(U_L) = h(V)h(U_L/V) = 1$. Again, $L^\times/U_L \simeq \mathbf{Z}$. So $h(L^\times) = h(\mathbf{Z})h(U_L)$. Now $h(U_L) = 1$ and $h(\mathbf{Z}) = n$, since $\hat{H}^0(G, \mathbf{Z})$ has order n and $H^1(G, \mathbf{Z})$ is trivial. Hence $h(L^\times) = n$. $\square$

Corollary 1.13. — *Let L/K be a cyclic extension of degree n . Then $H^2(L/K)$ is of order $n = [L : K]$.*

Démonstration. — We have

$$h(L^\times) = \frac{\text{Card } H^2(G, L^\times)}{\text{Card } H^1(G, L^\times)}.$$

Now Corollary 1.12 gives $h(L^\times) = n$. Moreover, $H^1(G, L^\times) = 0$ by Hilbert Theorem 90. Hence $\text{Card } H^2(G, L^\times) = n$. But $H^2(G, L^\times)$ is $H^2(L/K)$, whence the result. $\square$

1.5. An Ugly Lemma. —

Lemma 1.14. — *Let G be a finite group and let M be a G -module and suppose that p, q are integers with $p \geq 0, q \geq 0$. Assume that:*

- (a) $H^i(H, M) = 0$ for all $0 < i < q$ and all subgroups H of G ;
- (b) if $H \subset K \subset G$, with H invariant in K and K/H cyclic of prime order, then the order of $H^q(K/H, M^H)$, respectively $\hat{H}^0(K/H, M^H)$ if $q = 0$, divides $(K : H)^p$.

Then the same is true of G . That is, $H^q(G, M)$, respectively $\hat{H}^0(G, M)$ if $q = 0$, is of order dividing $(G : 1)^p$.

Démonstration. — Since the restriction map

$$\text{Res} : H^q(G, M) \longrightarrow H^q(G_p, M)$$

is injective on the p -primary components of $H^q(G, M)$, where G_p denotes a Sylow p -subgroup of G , we may confine our attention to the case in which G is a p -group. We now argue by induction on the order of G .

Assume that G has order greater than 1. Choose a subgroup H of G which is invariant and of index p . We apply the induction hypothesis to H .

We know from (b) that, for $q > 0$, the order of $H^q(G/H, M^H)$ divides $(G : H)^p = p^p$ and, by the induction hypothesis, $H^q(H, M)$ has order dividing $(H : 1)^p$. Now it follows from (a) that we have an exact sequence (Chapter IV, § 5)

$$0 \longrightarrow H^q(G/H, M^H) \xrightarrow{\text{Inf}} H^q(G, M) \xrightarrow{\text{Res}} H^q(H, M).$$

Thus $H^q(G, M)$ has order dividing $p^p \cdot (H : 1)^p = (G : 1)^p$.

For $q = 0$, we recall (see Chapter IV, § 6) that

$$\hat{H}^0(G, M) = M^G / N_G M.$$

Then we have the exact sequence

$$M^H / N_H M \xrightarrow{N_{G/H}} M^G / N_G M \longrightarrow (M^H)^{G/H} / N_{G/H} M^H,$$

where $N_{G/H}$ denotes the norm map and the second map is induced by the identity. The remainder of the argument now runs as before. $\square$

1.6. End of Proofs. —

Proposition 1.15. — *Let L/K be a finite galois extension with Galois group G of order $n = [L : K]$. Then $H^2(L/K)$ is cyclic of order n , generated by an element $u_{L/K} \in H^2(K_{\text{nr}}/K)$ such that $\text{inv}_K(u_{L/K}) = 1/n$.*

Démonstration. — In Lemma 1.14, take $M = L^\times$, $p = 1$ and $q = 2$. Condition (a) is satisfied by “Theorem 90” and (b) is true by Proposition 1.11 and Corollary 1.12. Hence $H^2(G, L^\times)$ has order dividing $(G : 1) = n$. But by Proposition 1.8 and Corollary 1.9, $H^2(L/K)$ contains a cyclic subgroup of order n , generated by $u_{L/K} \in H^2(K_{\text{nr}}/K)$ and such that $\text{inv}_K(u_{L/K}) = 1/n$. It follows that this subgroup is all of $H^2(L/K)$, whence Proposition 1.15.

It follows from Proposition 1.15 that $H^2(L/K)$ is contained in $H^2(K_{\text{nr}}/K)$.

We turn now to the proof of Theorem 1. The theorem asserts that the inclusion $H^2(K_{\text{nr}}/K) \subset \text{Br}(K)$ is actually equality. Now by definition,

$$\text{Br}(K) = \bigcup_L H^2(L/K),$$

where L runs through the set of finite galois extensions of K . But as remarked above, $H^2(L/K) \subset H^2(K_{\text{nr}}/K)$. Hence $\text{Br}(K) = H^2(K_{\text{nr}}/K)$, as was to be proved.

Evidently, Theorem 1.3 follows from Theorem 1.1 and Proposition 1.8. $\square$

1.7. An Auxiliary Result. — We have now proved all the statements in § 1.1 and we conclude the present chapter with a result which has applications to global fields.

Let A be an abelian group and let n be an integer ≥ 1 . Consider the cyclic group $\mathbf{Z}/n\mathbf{Z}$ with trivial action on A . We shall denote the corresponding Herbrand quotient by $h_n(A)$, whenever it is defined. We have

$$h_n(A) = \frac{\text{Order}(A/nA)}{\text{Order}({}_nA)}$$

where ${}_nA$ is the set of $a \in A$ such that $na = 0$. (Alternatively, we could begin with the map $A \xrightarrow{n} A$ and take $h_n(A)$ to be

$$\text{order}(\text{Coker}(n)) / \text{order}(\text{Ker}(n)).$$

Now let K be a local field. Then for $\alpha \in K$ there is a normalized absolute value, denoted by $|\alpha|_K$ (see Chapter II, § 11). If $\alpha \in \mathcal{O}_K$, then

$$|\alpha|_K = 1 / \text{Card}(\mathcal{O}_K / \alpha \mathcal{O}_K).$$

Proposition 1.16. — *Let K be a local field and let $n \geq 1$ be an integer prime to the characteristic of K . Then*

$$h_n(K^\times) = n / |n|_K.$$

Démonstration. — Suppose that K has characteristic 0. We have $h_n(K^\times) = h_n(\mathbf{Z}) \cdot h_n(U_K)$. Now $h_n(\mathbf{Z}) = n$; so we must compute $h_n(U_K)$. As in Proposition 1.11, we consider a subgroup V of U_K which is open and isomorphic to the additive group of $\mathcal{O}_K$. We have $h_n(U_K) = h_n(V) \cdot h_n(U_K/V)$ and since U_K/V is finite, $h_n(U_K/V) = 1$. We have

$$h_n(V) = h_n(\mathcal{O}_K)$$

and

$$h_n(\mathcal{O}_K) = \text{Card}(\mathcal{O}_K / n \mathcal{O}_K) = 1 / |n|_K.$$

Whence

$$h_n(K^\times) = n \cdot (1 / |n|_K) = n / |n|_K.$$

Suppose now that K has characteristic p . We take the same steps as before. First, $h_n(K^\times) = n \cdot h_n(U_K)$. Now consider the exact sequence

$$0 \longrightarrow U_K^1 \longrightarrow U_K \longrightarrow k^\times \longrightarrow 0$$

where U_K^1 is a pro- p -group (cf. Lemma 1.5). Since n is prime to p it follows that $h_n(U_K^1) = 1$ and that $h_n(k^\times) = 1$. So $h_n(U_K) = 1$. Whence the result. $\square$

We note that the statement of the proposition is also correct for $\mathbf{R}$ or $\mathbf{C}$. In these cases we have $|n|_{\mathbf{R}} = |n|$, $|n|_{\mathbf{C}} = |n|^2$ and one can check directly that, for $\mathbf{R}$, $h_n(\mathbf{R}^\times) = n / |n| = 1$ and, for $\mathbf{C}$, $h_n(\mathbf{C}^\times) = n / |n|_{\mathbf{C}} = 1/n$.

Appendix. —

Division Algebras Over a Local Field. — It is known that elements of Brauer groups correspond to skew fields (cf., for instance, “Séminaire Cartan”, 1950/51, Exposés 6/7), and we are going to use this correspondence to give a description of skew fields and the corresponding invariants. Most results will be stated without proof.

Let K be a local field and let D be a division algebra over K , with centre K and $[D : K] = n^2$. The valuation v of K extends in a unique way from K to D (for example, by extending first to $K(\alpha)$, $\alpha \in D$, and then fitting the resulting extensions together). The field D is complete with respect to this valuation and, in an obvious notation, $\mathcal{O}_D$ is of degree n^2 over $\mathcal{O}_K$. Let d be the residue field of D ; we have $n^2 = ef$ where e is the ramification index and $f = [d : k]$.

Now $e \leq n$; for there exists $\alpha \in D$ such that $v_D(\alpha) = e^{-1}$ and α belongs to a commutative subfield of degree at most n over K . The residue field d is commutative, since k is a finite field, and $d = k(\bar{\alpha})$ for some $\alpha \in D$. Hence $f \leq n$. Together with $n^2 = ef$, the inequalities $e \leq n$ and $f \leq n$ yield $e = n$ and $f = n$.

Since $[d : k] = n$, we can find $\bar{\alpha} \in d$ such that $k(\bar{\alpha}) = d$. Now choose a corresponding $\alpha \in \mathcal{O}_D$ and let $L = K(\alpha)$. Evidently $[L : K] \leq n$, since L is a commutative subfield of D . On the other hand, $\bar{\alpha}$ is an element of l (the residue field of L) and $l = d$; hence $[l : k] = n$. It follows that $[L : K] = n$ and L is unramified. We state this last conclusion as: D contains a maximal commutative subfield L which is unramified over K .

The element $\delta \in \text{Br}(K)$ corresponding to D splits in L , that is $\delta \in H^2(L/K)$.

So any element in $\text{Br}(K)$ is split by an unramified extension and we have obtained a new proof of Theorem 1.1.

Description of the Invariant. — The extension L of K constructed above is not unique, but the Skolem–Noether theorem (Bourbaki, “Algèbre”, Chap. 8, § 10) shows that all such extensions are conjugate. The same theorem shows that any automorphism of L is induced by an inner automorphism of D . Hence there exists $\gamma \in D$ such that $\gamma L \gamma^{-1} = L$ and the inner automorphism $x \mapsto \gamma x \gamma^{-1}$ on L is the Frobenius F . Moreover γ is determined, up to multiplication by an element of $L^\times$.

Let v_L be the valuation $v_L : L^\times \rightarrow \mathbf{Z}$ of L ; so that $v_D : D^\times \rightarrow (1/n)\mathbf{Z}$ extends v_L . The image $i(D)$ of $v_D(\gamma)$ in $(1/n)\mathbf{Z}/\mathbf{Z} = \mathbf{Q}/\mathbf{Z}$ is independent

of the choice of γ . One can prove that $i(D) = \text{inv}_K(\delta)$, where $\delta \in \text{Br}(K)$ is associated with D .

We can express the definition of $i(D)$ in a slightly different way. The map $x \mapsto \gamma^n x \gamma^{-n}$ is equal to F^n on L and so is the identity. It follows that γ^n commutes with L and $\gamma^n = c \in L^\times$. Now

$$v_D(\gamma) = \frac{1}{n} v_D(\gamma^n) = \frac{1}{n} v_D(c) = \frac{1}{n} v_L(c).$$

Hence we have $v_D(\gamma) = (1/n)v_L(c) = i/n$ where $c = \pi_L^i u$.

Application. — Suppose that K'/K is an extension of degree n . By Theorem 1.3, Corollary 2, an element $\delta \in \text{Br}(K)$ is killed by K' . Hence: any extension K'/K of degree n can be embedded in D as a maximal commutative subfield. This may be stated more spectacularly as: any irreducible equation of degree n over K can be solved in D .

Exercise. — Consider the 2-adic field $\mathbf{Q}_2$ and let H be the quaternion skew field over $\mathbf{Q}_2$. Prove that the ring of integers in H consists of the elements $a + bi + cj + dk$ where $a, b, c, d \in \mathbf{Z}_2$, or $a, b, c, d \equiv \frac{1}{2} \pmod{\mathbf{Z}_2}$. Make a list of the seven (up to conjugacy) quadratic subfields of H .

Solution. — Write $H = \mathbf{Q}_2 + \mathbf{Q}_2 i + \mathbf{Q}_2 j + \mathbf{Q}_2 k$, with

$$i^2 = j^2 = -1, \quad ij = k = -ji.$$

For $x = a + bi + cj + dk$, the reduced trace and norm are

$$\text{Trd}(x) = 2a, \quad \text{Nrd}(x) = a^2 + b^2 + c^2 + d^2.$$

An element of the division algebra is integral if and only if its reduced characteristic polynomial has coefficients in $\mathbf{Z}_2$, or equivalently if $\text{Trd}(x)$ and $\text{Nrd}(x)$ are in $\mathbf{Z}_2$.

It is immediate that this holds if $a, b, c, d \in \mathbf{Z}_2$. It also holds if $a, b, c, d \equiv \frac{1}{2} \pmod{\mathbf{Z}_2}$: then $2a \in \mathbf{Z}_2$, and if $2a, 2b, 2c, 2d$ are all odd, the sum of their squares is congruent to 4 (mod 8), so

$$a^2 + b^2 + c^2 + d^2 = \frac{(2a)^2 + (2b)^2 + (2c)^2 + (2d)^2}{4}$$

belongs to $\mathbf{Z}_2$.

Conversely, suppose x is integral. Multiplying the four coordinates by a common power of 2, choose $r \geq 0$ so that

$$2^r a, 2^r b, 2^r c, 2^r d \in \mathbf{Z}_2$$

and not all of these four elements are divisible by 2. If $r = 0$, then all four coordinates are in $\mathbf{Z}_2$. If $r > 0$, the integrality of the norm implies that the sum of the four squares

$$(2^r a)^2 + (2^r b)^2 + (2^r c)^2 + (2^r d)^2$$

is divisible by 2^{2r} . Modulo 4, a square is either 0 or 1; therefore the number of odd terms must be a multiple of 4. Since at least one term is odd, all four are odd. Their squares then sum to 4 (mod 8), so the valuation of the sum is exactly 2. Hence $r = 1$, and

$$a, b, c, d \equiv \frac{1}{2} \pmod{\mathbf{Z}_2}.$$

This proves the asserted description of the ring of integers.

The quadratic extensions of $\mathbf{Q}_2$ are classified by the non-trivial square classes in $\mathbf{Q}_2^\times / \mathbf{Q}_2^{\times 2}$. Since

$$\mathbf{Q}_2^\times / \mathbf{Q}_2^{\times 2} = \{1, -1, 2, -2, 3, -3, 6, -6\},$$

the seven quadratic subfields, up to conjugacy, may be listed as

$$\begin{aligned} &\mathbf{Q}_2(\sqrt{-1}), \quad \mathbf{Q}_2(\sqrt{2}), \quad \mathbf{Q}_2(\sqrt{-2}), \quad \mathbf{Q}_2(\sqrt{3}), \\ &\mathbf{Q}_2(\sqrt{-3}), \quad \mathbf{Q}_2(\sqrt{6}), \quad \mathbf{Q}_2(\sqrt{-6}). \end{aligned}$$

Each embeds in H as a maximal commutative subfield, and by Skolem–Noether any two embeddings of the same quadratic field are conjugate in $H^\times$. $\square$

2. Abelian Extensions of Local Fields

2.1. Cohomological Properties. — Let L/K be a finite galois extension of local fields with Galois group $G = G(L/K)$ of order n . We have seen (§ 1.1, Theorem 3, Cor. 2) that the group $H^2(L/K) = H^2(G, L^\times)$ is cyclic of order n and contains a generator $u_{L/K}$ such that $\text{inv}_K(u_{L/K}) = 1/n \in \mathbf{Q}/\mathbf{Z}$. On the other hand, we know that $H^1(G, L^\times) = 0$.

Now let H be a subgroup of G of order m . Since H is the Galois group of L/K' for some $K' \supset K$, we also have $H^1(H, L^\times) = 0$ and $H^2(H, L^\times)$ is cyclic of order m and generated by $u_{L/K'}$. To go further, we need to know more about $u_{L/K}$. Now we have the restriction map $\text{Res} : \text{Br}(K) \rightarrow \text{Br}(K')$ and this suggests that $u_{L/K'} = \text{Res}(u_{L/K})$.

To see that this is the case, we simply check on invariants. We have

$$\begin{aligned} \text{inv}_{K'}(\text{Res } u_{L/K}) &= [K' : K] \text{inv}_K(u_{L/K}) \\ &= [K' : K]/[L : K] = 1/[L : K'] \\ &= \text{inv}_{K'}(u_{L/K'}). \end{aligned}$$

We can now apply Tate's theorem (Chapter IV, § 10) to obtain:

Theorem 2.1. — *For all $q \in \mathbf{Z}$, the map $\alpha \mapsto \alpha \cdot u_{L/K}$ given by the cup-product is an isomorphism of $\hat{H}^q(G, \mathbf{Z})$ onto $\hat{H}^{q+2}(G, L^\times)$.*

A similar statement holds if H is a subgroup of G corresponding to an extension L/K' . The mappings Res and Cor connect the two isomorphisms and we have a more explicit statement in terms of diagrams.

STATEMENT. The diagrams

$$\begin{array}{ccc} \hat{H}^q(G, \mathbf{Z}) & \xrightarrow{u_{L/K}} & \hat{H}^{q+2}(G, L^\times) \\ \text{Res} \downarrow & & \downarrow \text{Res} \\ \hat{H}^q(H, \mathbf{Z}) & \xrightarrow{u_{L/K'}} & \hat{H}^{q+2}(H, L^\times) \end{array} \quad \begin{array}{ccc} \hat{H}^q(G, \mathbf{Z}) & \xrightarrow{u_{L/K}} & \hat{H}^{q+2}(G, L^\times) \\ \text{Cor} \uparrow & & \uparrow \text{Cor} \\ \hat{H}^q(H, \mathbf{Z}) & \xrightarrow{u_{L/K'}} & \hat{H}^{q+2}(H, L^\times) \end{array}$$

are commutative.

Démonstration. — As above, $u_{L/K'} = \text{Res}(u_{L/K})$. We must show that

$$\text{Res}_{K/K'}(u_{L/K} \cdot \alpha) = u_{L/K'} \cdot \text{Res}_{K/K'}(\alpha).$$

The left-hand side is $\text{Res}_{K/K'}(u_{L/K}) \cdot \text{Res}_{K/K'}(\alpha)$ (see Cartan–Eilenberg, “Homological Algebra”, Chap. XII, p. 256) and so commutativity with Res is proved.

For the second diagram we have to show that $\text{Cor}(u_{L/K'} \cdot \beta) = u_{L/K} \cdot \text{Cor}(\beta)$. Now

$$\text{Cor}(u_{L/K'} \cdot \beta) = \text{Cor}(\text{Res}(u_{L/K}) \cdot \beta) = u_{L/K} \cdot \text{Cor}(\beta)$$

(Cartan–Eilenberg, loc. cit.) and this proves the commutativity of the second diagram. $\square$

2.2. The Reciprocity Map. — We shall be particularly concerned with the case $q = -2$ of the foregoing discussion. By definition $\hat{H}^{-2}(G, \mathbf{Z})$ is $H_1(G, \mathbf{Z})$ and we know that $H_1(G, \mathbf{Z}) = G/G' = G^{\text{ab}}$. On the other hand, $\hat{H}^0(L/K) = K^\times/N_{L/K}L^\times$, where $N_{L/K}$ denotes the norm. In this case, Theorem 2.1 reads as follows.

Theorem 2.2. — *The cup-product by $u_{L/K}$ defines an isomorphism of $G^{\text{ab}}(L/K)$ onto $K^\times/N_{L/K}L^\times$.*

We give a name to the isomorphism just constructed, or rather to its inverse. Define $\theta = \theta_{L/K}$ to be the isomorphism of $K^\times/N_{L/K}L^\times$ onto G^{ab} , which is inverse to the cup-product by $u_{L/K}$. The map θ is called the local reciprocity map or the norm residue symbol.

If $a \in K^\times$ corresponds to $\bar{a} \in K^\times/N_{L/K}L^\times$, then we write $\theta_{L/K}(\bar{a}) = (a, L/K)$.

The norm residue symbol is so named since it tells whether or not $a \in K^\times$ is a norm from $L^\times$. Namely, $(a, L/K) = 0$ (remember that 0 means 1!) if and only if a is a norm from $L^\times$.

Observe that if L/K is abelian, then $G^{\text{ab}} = G$ and we have an isomorphism

$$\theta : K^\times/N_{L/K}L^\times \longrightarrow G.$$

2.3. Characterization of $(a, L/K)$ by Characters. — Let L/K be a galois extension with group G . We start from an $a \in K^\times$ and we seek a characterization of $(a, L/K) \in G^{\text{ab}}$. For ease of writing we set $s_a = (a, L/K)$. Let $\chi \in \text{Hom}(G, \mathbf{Q}/\mathbf{Z}) = H^1(G, \mathbf{Q}/\mathbf{Z})$ be a character of degree 1 of G and let $\delta\chi \in H^2(G, \mathbf{Z})$ be the image of χ by the coboundary map $\delta : H^1(G, \mathbf{Q}/\mathbf{Z}) \rightarrow H^2(G, \mathbf{Z})$ (cf. § 1.1). Let $\bar{a} \in K^\times/N_{L/K}L^\times = \hat{H}^0(G, L^\times)$ be the image of a . The cup-product $\bar{a} \cdot \delta\chi$ is an element of $H^2(G, L^\times) \subset \text{Br}(K)$.

Proposition 2.3. — *With the foregoing notation, we have the formula*

$$\chi(s_a) = \text{inv}_K(\bar{a} \cdot \delta\chi).$$

Démonstration. — By definition $s_a \cdot u_{L/K} = \bar{a} \in \hat{H}^0(G, L^\times)$, s_a being identified with an element of $\hat{H}^{-2}(G, \mathbf{Z})$. Using the associativity of the cup-product, this gives

$$\bar{a} \cdot \delta\chi = u_{L/K} \cdot s_a \cdot \delta\chi = u_{L/K} \cdot (s_a \cdot \delta\chi) = u_{L/K} \cdot \delta(s_a \cdot \chi)$$

with $s_a \cdot \chi \in \hat{H}^{-1}(G, \mathbf{Q}/\mathbf{Z})$.

Now

$$\hat{H}^{-1}(G, \mathbf{Q}/\mathbf{Z}) \xrightarrow{\delta} \hat{H}^0(G, \mathbf{Z}) = \mathbf{Z}/n\mathbf{Z}$$

and we identify $\hat{H}^{-1}(G, \mathbf{Q}/\mathbf{Z})$ with $\mathbf{Z}/n\mathbf{Z}$. Moreover, the identification between $\hat{H}^{-2}(G, \mathbf{Z})$ and G^{ab} has been so made in order to ensure that $s_a \cdot \chi = \chi(s_a)$ (see “Corps Locaux”, Chap. XI, Annexe pp. 184–186). Write $s_a \cdot \chi = r/n$, $r \in \mathbf{Z}$. Then $\delta(r/n) \in \hat{H}^0(G, \mathbf{Z})$ and $\delta(r/n) = r$. Hence $u_{L/K} \cdot (s_a \cdot \delta\chi) = r \cdot u_{L/K}$ and the invariant of this cohomology class is just $r/n = \chi(s_a)$. So the proposition is proved. $\square$

As an application we consider the following situation. Consider a tower of galois extensions $K \subset L' \subset L$ with $G = G(L/K)$ and $H = G(L/L')$. Then, if χ' is a character of $(G/H)^{\text{ab}}$ and χ is the corresponding character of G^{ab} , and if $a \in K^\times$ induces $s_a \in G^{\text{ab}}$ and $s'_a \in (G/H)^{\text{ab}}$ under the natural map $s_a \mapsto s'_a$, we have

$$\chi(s_a) = \chi'(s'_a).$$

This follows from Proposition 2.3 and the fact that the inflation map transforms χ' (resp. $\delta\chi'$) into χ (resp. $\delta\chi$).

This compatibility allows us to define s_a for any abelian extension; in particular, taking $L = K^{\text{ab}}$, the maximal abelian extension of K , we get a homomorphism

$$\theta_K : K^\times \longrightarrow G(K^{\text{ab}}/K)$$

defined by $a \mapsto (a, K^{\text{ab}}/K)$.

2.4. Variations with the Fields Involved. — Having considered the effect on $(a, L/K)$ of extensions of L we turn now to consider extensions of K . Let K'/K be a separable extension and let $K^{\text{ab}}, K'^{\text{ab}}$ be the maximal abelian extensions of K, K' respectively.

We look at the first of the diagrams in the Statement of § 2.1 and the case $q = -2$. Taking the projective limit of the groups involved, we obtain a commutative diagram:

$$\begin{array}{ccc} K^\times & \xrightarrow{\theta_K} & G_K^{\text{ab}} \\ \text{incl} \downarrow & & \downarrow V \\ K'^\times & \xrightarrow{\theta_{K'}} & G_{K'}^{\text{ab}} \end{array}$$

Here V denotes the transfer (Chapter IV, § 6), $G_{K'}^{\text{ab}}$ denotes $G(K'^{\text{ab}}/K')$ and G_K^{ab} denotes $G(K^{\text{ab}}/K)$.

Similarly, using the second of the diagrams in the Statement, we obtain a commutative diagram:

$$\begin{array}{ccc} K'^\times & \xrightarrow{\theta_{K'}} & G_{K'}^{\text{ab}} \\ N_{K'/K} \downarrow & & \downarrow i \\ K^\times & \xrightarrow{\theta_K} & G_K^{\text{ab}} \end{array}$$

where i is induced by the inclusion of $G_{K'}$ into G_K .

[Note that if K'/K is an inseparable extension, then in the first of these diagrams the transfer, V , should be replaced by qV where q is the inseparable factor of the degree of the extension K'/K . The second diagram holds even in the inseparable case.]

2.5. Unramified Extensions. — In this case it is possible to compute the norm residue symbol explicitly in terms of the Frobenius element:

Proposition 2.4. — *Let L/K be an unramified extension of degree n and let $F \in G(L/K)$ be the Frobenius element. Let $a \in K^\times$ and let $v(a) \in \mathbf{Z}$ be its normalized valuation. Then $(a, L/K) = F^{v(a)}$.*

Démonstration. — Let χ be an element of $\text{Hom}(G(L/K), \mathbf{Q}/\mathbf{Z})$. By Proposition 2.3, we have

$$\chi((a, L/K)) = \text{inv}_K(\bar{a} \cdot \delta\chi).$$

The map $\text{inv}_K : H^2(G(L/K), L^\times) \rightarrow \mathbf{Q}/\mathbf{Z}$ has been defined as a composition

$$H^2(G(L/K), L^\times) \xrightarrow{v} H^2(G(L/K), \mathbf{Z}) \xrightarrow{\delta^{-1}} H^1(G(L/K), \mathbf{Q}/\mathbf{Z}) \xrightarrow{\gamma} \mathbf{Q}/\mathbf{Z}.$$

We have $v(\bar{a} \cdot \delta\chi) = v(a) \cdot \delta\chi$, hence

$$\text{inv}_K(\bar{a} \cdot \delta\chi) = \gamma \circ \delta^{-1} \circ v(\bar{a} \cdot \delta\chi) = v(a)\gamma(\chi) = v(a)\chi(F) = \chi(F^{v(a)}).$$

This shows that, for any character χ of $G(L/K)$, $(a, L/K) = F^{v(a)}$. $\square$

Corollary 2.5. — *Let E/K be a finite abelian extension. The norm residue symbol $K^\times \rightarrow G(E/K)$ maps U_K onto the inertia subgroup T of $G(E/K)$.*

Démonstration. — Let L be the sub-extension of E corresponding to T . By Proposition 2.4, the image of U_K in $G(L/K)$ is trivial; this means that the image of U_K in $G(E/K)$ is contained in T . Conversely, let $t \in T$, and let $f = [L : K]$; there exists $a \in K^\times$ such that $t = (a, E/K)$. Since $t \in T$, Proposition 2.4 shows that f divides $v(a)$; hence, there exists $b \in E^\times$ such that $v(a) = v(Nb)$. If we put $u = a \cdot (Nb)^{-1}$, we have $u \in U_K$ and $(u, E/K) = (a, E/K) = t$. $\square$

2.6. Norm Subgroups. —

Definition 2.6. — A subgroup M of $K^\times$ is called a norm subgroup if there exists a finite abelian extension L/K with $M = N_{L/K}L^\times$.

Example. Let $m \geq 1$ be an integer, and let M_m be the set of elements $a \in K^\times$ with $v_K(a) \equiv 0 \pmod{m}$; it follows from Proposition 2.4 (or from a direct computation of norms) that M_m is the norm group of the unramified extension of K of degree m .

Norm subgroups are closely related to the reciprocity map

$$\theta_K : K^\times \longrightarrow G_K^{\text{ab}} = G(K^{\text{ab}}/K)$$

defined in § 2.3. By construction, θ_K is obtained by projective limit from the isomorphisms $K^\times/N_{L/K}L^\times \rightarrow G_{L/K}$, where L runs through all finite abelian extensions of K . If we put

$$\tilde{K} = \varprojlim_L K^\times/N_{L/K}L^\times,$$

we see that θ_K can be factored into

$$K^\times \xrightarrow{i} \tilde{K} \xrightarrow{\tilde{\theta}} G_K^{\text{ab}},$$

where i is the natural map, and $\tilde{\theta}$ is an isomorphism. Note that $\tilde{K}$ is just the completion of $K^\times$ with respect to the topology defined by the norm subgroups.

This shows that norm subgroups of $K^\times$ and open subgroups of G_K^{ab} correspond to each other in a one-one way: if U is an open subgroup of G_K^{ab} , with fixed field L , we attach to U the norm subgroup $\theta_K^{-1}(U) = N_{L/K}L^\times$; if M is a norm subgroup of $K^\times$, we attach to it the adherence of $\theta_K(M)$; the corresponding field L_M is then the set of elements in K^{ab} which are invariant by the $\theta_K(a)$, for $a \in M$. We thus get a “Galois correspondence” between norm subgroups and finite abelian extensions; we state it as a proposition:

Proposition 2.7. — (a) *The map $L \mapsto N_{L/K}L^\times$ is a bijection of the set of finite abelian extensions of K onto the set of norm subgroups of $K^\times$.*

(b) *This bijection reverses the inclusion.*

(c)

$$N_{LL'/K}(LL')^\times = N_{L/K}L^\times \cap N_{L'/K}L'^\times$$

and

$$N_{L \cap L'/K}(L \cap L')^\times = N_{L/K}L^\times \cdot N_{L'/K}L'^\times.$$

(d) *Any subgroup of $K^\times$ which contains a norm subgroup is a norm subgroup.*

(For a direct proof, see “Corps Locaux”, Chap. XI, § 4.) Non-abelian extensions give the same norm subgroups as the abelian ones:

Proposition 2.8. — *Let E/K be a finite extension, and let L/K be the largest abelian extension contained in E . Then we have*

$$N_{E/K}E^\times = N_{L/K}L^\times.$$

Démonstration. — This follows easily from the properties of the norm residue symbol proved in § 2.4; for more details, see Artin–Tate, “Class Field Theory”, pp. 228–229, or “Corps Locaux”, p. 180. (These two books give only the case

where E/K is separable; the general case reduces to this one by observing that $NL = K$ when L is a purely inseparable extension of K .) $\square$

Corollary 2.9 (Limitation theorem). — *The index $(K^\times : N_{E/K}E^\times)$ divides $[E : K]$. It is equal to $[E : K]$ if and only if E/K is abelian.*

Démonstration. — This follows from the fact that the index of $N_{L/K}L^\times$ in $K^\times$ is equal to $[L : K]$. $\square$

2.7. Statement of the Existence Theorem. — It gives a characterization of the norm subgroups of $K^\times$:

Theorem 2.10. — *A subgroup M of $K^\times$ is a norm subgroup if and only if it satisfies the following two conditions:*

- (1) *its index $(K^\times : M)$ is finite;*
- (2) *M is open in $K^\times$.*

(Note that, if (1) is satisfied, (2) is equivalent to “ M is closed”.)

Proof of necessity. If $M = NL^\times$, where L is a finite abelian extension of K , we know that $K^\times/M$ is isomorphic to $G_{L/K}$; hence $(K^\times : M)$ is finite. Moreover, one checks immediately that $N : L^\times \rightarrow K^\times$ is continuous and proper (the inverse image of a compact set is compact); hence $M = NL^\times$ is closed, cf. Bourbaki, “Top. Gén.”, Chap. I, § 10. As remarked above, this shows that M is open. [This last property of the norm subgroups may also be expressed by saying that the reciprocity map

$$\theta_K : K^\times \longrightarrow G_K^{\text{ab}}$$

is continuous.]

Proof of sufficiency. See § 3.8, where we shall deduce it from Lubin–Tate’s theory. The usual proof, reproduced for instance in “Corps Locaux”, uses Kummer and Artin–Schreier equations.

We give now some equivalent formulations.

Consider the reciprocity map $\theta_K : K^\times \rightarrow G_K^{\text{ab}}$. By Proposition 2.4, the composition

$$K^\times \xrightarrow{\theta_K} G_K^{\text{ab}} \longrightarrow G(K_{\text{nr}}/K) = \widehat{\mathbf{Z}}$$

is just the valuation map $v : K^\times \rightarrow \mathbf{Z}$. Hence we have a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & U_K & \longrightarrow & K^\times & \xrightarrow{v} & \mathbf{Z} \longrightarrow 0 \\ & & \downarrow \theta & & \downarrow \theta_K & & \downarrow \text{id} \\ 0 & \longrightarrow & I_K & \longrightarrow & G_K^{\text{ab}} & \longrightarrow & \widehat{\mathbf{Z}} \longrightarrow 0, \end{array}$$

where $I_K = G(K^{\text{ab}}/K_{\text{nr}})$ is the inertia subgroup of G_K^{ab} , and $G(K_{\text{nr}}/K)$ is identified with $\widehat{\mathbf{Z}}$.

The map $\theta : U_K \rightarrow I_K$ is continuous, and its image is dense (cf. Corollary 2.5); since U_K is compact, it follows that it is surjective.

We can now state two equivalent formulations of the existence theorem.

Theorem 2.11. — *The map $\theta : U_K \rightarrow I_K$ is an isomorphism.*

Theorem 2.12. — *The topology induced on U_K by the norm subgroups is the natural topology of U_K .*

The group I_K is just

$$\varprojlim_M U_K / (M \cap U_K),$$

where M runs through all norm subgroups of $K^\times$; the equivalence of Theorem 2.11 and Theorem 2.12 follows from this and a compactness argument. The fact that Theorem 2.10 implies Theorem 2.12 is clear; the converse is easy, using Proposition 2.4.

Corollary 2.13. — *The exact sequence*

$$0 \longrightarrow U_K \longrightarrow K^\times \longrightarrow \mathbf{Z} \longrightarrow 0$$

gives by completion the exact sequence

$$0 \longrightarrow U_K \longrightarrow \widetilde{K} \longrightarrow \widehat{\mathbf{Z}} \longrightarrow 0.$$

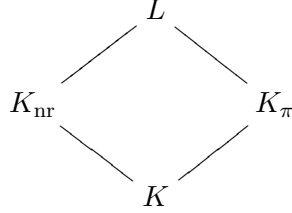
Loosely speaking, this means that $\widetilde{K}$ is obtained from $K^\times$ by “replacing $\mathbf{Z}$ by $\widehat{\mathbf{Z}}$ ”.

2.8. Some Characterizations of $(a, L/K)$. — Let L be an abelian extension of K containing K_{nr} , the maximal unramified extension, and put $G = G_{L/K}$. We want to give characterizations of the reciprocity map $\theta : K^\times \rightarrow G_{L/K}$. Since $K_{\text{nr}} \subset L$, we have an exact sequence

$$0 \longrightarrow H \longrightarrow G_{L/K} \longrightarrow \widehat{\mathbf{Z}} \longrightarrow 0$$

where $H = G(L/K_{\text{nr}})$ and $\widehat{\mathbf{Z}}$ is identified with $G(K_{\text{nr}}/K)$. Choose a local uniformizer π in K and write $\sigma_\pi = \theta(\pi) = (\pi, L/K) \in G_{L/K}$. We know that σ_π maps onto the Frobenius element $F \in G_{K_{\text{nr}}/K}$. Moreover, we can write $G_{L/K}$ as a direct product of subgroups $G_{L/K} = H \cdot I_\pi$, where I_π is generated by σ_π . Corresponding to this we have $L = K_{\text{nr}} \otimes K_\pi$, where K_π is the fixed field of I_π . In terms of a diagram, the interrelationship between the fields is

expressed by



where K_{nr} and K_{π} are linearly disjoint.

Proposition 2.14. — *Let $f : K^{\times} \rightarrow G$ be a homomorphism and assume that:*

- (1) *the composition $K^{\times} \xrightarrow{f} G \rightarrow G(K_{\text{nr}}/K)$, where $G \rightarrow G(K_{\text{nr}}/K)$ is the natural map, is the valuation map $v : K^{\times} \rightarrow \mathbf{Z}$;*
- (2) *for any uniformizing element $\pi \in K$, $f(\pi)$ is the identity on the corresponding extension K_{π} .*

Then f is equal to the reciprocity map θ .

Démonstration. — Note that condition (1) can be restated as: for $\alpha \in K^{\times}$, $f(\alpha)$ induces on K_{nr} the power of the Frobenius element $F^{v(\alpha)}$. We know that $f(\pi)$ is F on K_{nr} and that $\theta(\pi)$ is F on K_{nr} . On the other hand, $f(\pi)$ is 1 on K_{π} and $\theta(\pi)$ is 1 on K_{π} . Hence $f(\pi) = \theta(\pi)$ on L .

Now $K^{\times}$ is topologically generated by its uniformizing elements πu (write $\pi^n u$ as $(\pi u) \cdot \pi^{n-1}$). Hence $f = \theta$. $\square$

Proposition 2.15. — *Let $f : K^{\times} \rightarrow G$ be a homomorphism and assume that (1) of Proposition 2.14 holds, whilst (2) is replaced by:*

- (2') *if $\alpha \in K^{\times}$, if K'/K is a finite sub-extension of L and if α is a norm from $K'^{\times}$, then $f(\alpha)$ is trivial on K' .*

Then f is equal to the reciprocity map θ .

Démonstration. — It suffices to prove that (2') implies (2). That is, we have to prove that if π is a uniformizing element, then $f(\pi)$ is trivial on K_{π} . Let K'/K be a finite sub-extension of K_{π} . We want to prove that $\pi \in NK'^{\times}$. But $\theta(\pi)$ is trivial on K_{π} and so on K' . This implies $\pi \in NK'^{\times}$. $\square$

2.9. The Archimedean Case. — For global class-field theory it is necessary to extend these results to the (trivial) cases in which K is either $\mathbf{R}$ or $\mathbf{C}$. Let $G = G(\mathbf{C}/\mathbf{R})$. In the case $K = \mathbf{C}$, the Brauer group is trivial, $\text{Br}(\mathbf{C}) = 0$. On the other hand,

$$\text{Br}(\mathbf{R}) = H^2(G, \mathbf{C}^{\times}) = \mathbf{R}^{\times} / \mathbf{R}_{+}^{\times}$$

and so $\text{Br}(\mathbf{R})$ is of order 2.

The invariant $\text{inv}_{\mathbf{R}} : \text{Br}(\mathbf{R}) \rightarrow \mathbf{Q}/\mathbf{Z}$ has image $\{0, 1/2\}$ in $\mathbf{Q}/\mathbf{Z}$ and $\text{inv}_{\mathbf{C}} : \text{Br}(\mathbf{C}) \rightarrow \mathbf{Q}/\mathbf{Z}$ has image $\{0\}$. The group $H^2(G, \mathbf{C}^\times) = H^2(\mathbf{C}/\mathbf{R})$ is cyclic of order 2 and is generated by $u \in \text{Br}(\mathbf{R})$ such that $\text{inv}_{\mathbf{R}}(u) = 1/2$.

Under the reciprocity map (or rather its inverse) we have an isomorphism

$$G = H^{-2}(G, \mathbf{Z}) \longrightarrow H^0(G, \mathbf{C}^\times) = \mathbf{R}^\times / \mathbf{R}_+^\times.$$

3. Formal Multiplication in Local Fields

The results given in this chapter are due to Lubin–Tate, *Annals of Mathematics*, 81 (1965), 380–387.

For our purposes, the main consequences will be: (1) the construction of a cofinal system of abelian extensions of a given local field K ; (2) a formula giving $(\alpha, L/K)$ explicitly in such extensions; (3) the Existence Theorem of § 2.7.

In order to illustrate the ideas involved, we begin with the case $K = \mathbf{Q}_p$. The results to be proved were already known in this case (but were not easily obtained) and they will be shown to be trivial consequences of Lubin–Tate theory.

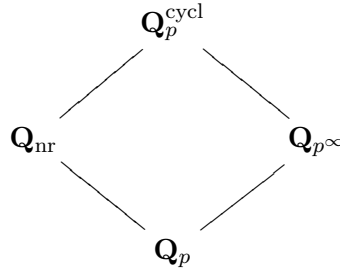
3.1. The Case $K = \mathbf{Q}_p$. —

Theorem 3.1. — *Let $\mathbf{Q}_p^{\text{cycl}}$ be the field generated over $\mathbf{Q}_p$ by all roots of unity. Then $\mathbf{Q}_p^{\text{cycl}}$ is the maximal abelian extension of $\mathbf{Q}_p$.*

In order to determine $(\alpha, L/K)$ it is convenient to split $\mathbf{Q}_p^{\text{cycl}}$ into parts. Define $\mathbf{Q}_{\text{nr}}$ to be the field generated over $\mathbf{Q}_p$ by roots of unity of order prime to p (so $\mathbf{Q}_{\text{nr}}$ is the maximal unramified extension of $\mathbf{Q}_p$) and define $\mathbf{Q}_{p^\infty}$ to be the field generated over $\mathbf{Q}_p$ by p^ν th roots of unity, $\nu = 1, 2, \dots$ (so $\mathbf{Q}_{p^\infty}$ is totally ramified). Then $\mathbf{Q}_{\text{nr}}$ and $\mathbf{Q}_{p^\infty}$ are linearly disjoint and

$$\mathbf{Q}_p^{\text{cycl}} = \mathbf{Q}_{\text{nr}} \cdot \mathbf{Q}_{p^\infty} = \mathbf{Q}_{\text{nr}} \otimes_{\mathbf{Q}_p} \mathbf{Q}_{p^\infty}.$$

We have a diagram:



Now $G(\mathbf{Q}_{\text{nr}}/\mathbf{Q}_p) = \widehat{\mathbf{Z}}$ and if $\sigma \in G(\mathbf{Q}_{p^\infty}/\mathbf{Q}_p)$ then σ is known by its action on the roots of unity. Let E be the group of p^ν th roots of unity, $\nu = 1, 2, \dots$. As an abelian group,

$$E \simeq \varinjlim \mathbf{Z}/p^\nu \mathbf{Z} = \mathbf{Q}_p/\mathbf{Z}_p.$$

We shall view E as a $\mathbf{Z}_p$ -module. There is a canonical map $\mathbf{Z}_p \rightarrow \text{End}(E)$, defined in an obvious way, and this map is an isomorphism. The action of the Galois group on E defines a homomorphism

$$G(\mathbf{Q}_{p^\infty}/\mathbf{Q}_p) \longrightarrow \text{Aut}(E) = U_p$$

and it is known that this is an isomorphism. (See Chapter III, and “Corps Locaux”, Chap. IX, § 4, and Chap. XIV, § 7.) If $u \in U_p$, we shall denote by $[u]$ the corresponding automorphism of $\mathbf{Q}_{p^\infty}/\mathbf{Q}_p$.

Theorem 3.2. — *If $\alpha = p^n \cdot u$ where $u \in U_p$, then $(\alpha, \mathbf{Q}_p^{\text{cycl}}/\mathbf{Q}_p) = \sigma_\alpha$ is described by:*

1. on $\mathbf{Q}_{\text{nr}}$, σ_α induces the n th power of the Frobenius automorphism;
2. on $\mathbf{Q}_{p^\infty}$, σ_α induces the automorphism $[u^{-1}]$.

Of these (1) is trivial and has already been proved in § 2.5, Proposition 2.4. The assertion (2) can be proved by (a) global methods, or (b) hard local methods (Dwork), or (c) Lubin–Tate theory (see § 3.4, Theorem 3.8).

Remark. Assertion (2) of Theorem 3.2 is equivalent to the following: if w is a primitive p^ν th root of unity and if $u \in U_p$, then

$$\sigma_u(w) = w^{u^{-1}} = 1 + \sum_{n=1}^{\infty} \binom{u^{-1}}{n} x^n,$$

where $w = 1 + x$.

3.2. Formal Groups. — The main game will be played with something which replaces the multiplicative group law $F(X, Y) = X + Y + XY$ and something instead of the binomial expansion. The group law will be a formal power series in two variables and we begin by studying such group laws.

Definition 3.3. — Let A be a commutative ring with 1 and let $F \in A[[X, Y]]$. We say that F is a commutative formal group law if:

- (a) $F(X, F(Y, Z)) = F(F(X, Y), Z)$;
- (b) $F(0, Y) = Y$ and $F(X, 0) = X$;
- (c) there is a unique $G(X)$ such that $F(X, G(X)) = 0$;
- (d) $F(X, Y) = F(Y, X)$;
- (e) $F(X, Y) \equiv X + Y \pmod{\deg 2}$.

In fact one can show that (c) and (e) are consequences of (a), (b) and (d). Here, two formal power series are said to be congruent modulo $\deg n$ if and only if they coincide in terms of degree strictly less than n .

Take $A = \mathcal{O}_K$. Let $F(X, Y)$ be a commutative formal group law defined over $\mathcal{O}_K$ and let $\mathfrak{m}_K$ be the maximal ideal of $\mathcal{O}_K$. If $x, y \in \mathfrak{m}_K$ then $F(x, y)$ converges and its sum $x * y$ belongs to $\mathfrak{m}_K$. Under this composition law, $\mathfrak{m}_K$ is a group which we denote by $F(\mathfrak{m}_K)$.

The same argument applies to an extension L/K and the maximal ideal $\mathfrak{m}_L$ in $\mathcal{O}_L$. We then obtain a group $F(\mathfrak{m}_L)$, defined for any algebraic extension of K by passage to the inductive limit from the finite case.

If $F(X, Y) = X + Y + XY$ then we recover the multiplicative group law of $1 + \mathfrak{m}_K$.

The elements of finite order of $F(\mathfrak{m}_{K_s})$ form a torsion group and $G(K_s/K)$ operates on this group. The structure of this Galois module presents an interesting problem which up to now has been solved only in special cases.

3.3. Lubin-Tate Formal Group Laws. — Let K be a local field, $q = \text{Card}(k)$ and choose a uniformizing element $\pi \in \mathcal{O}_K$. Let $\mathcal{F}_\pi$ be the set of formal power series f with:

- (1) $f(X) \equiv \pi X \pmod{\deg 2}$;
- (2) $f(X) \equiv X^q \pmod{\pi}$.

Two power series are said to be congruent modulo π if and only if each coefficient of their difference is divisible by π . So the second condition means that if we go to the residue field and denote by $\bar{f}(X)$ the corresponding element of $k[[X]]$, then $\bar{f}(X) = X^q$.

Examples:

- (a) $f(X) = \pi X + X^q$;
- (b) $K = \mathbf{Q}_p$, $\pi = p$,

$$f(X) = pX + \binom{p}{2}X^2 + \cdots + pX^{p-1} + X^p.$$

The following four propositions will be proved in § 3.5 as consequences of Proposition 3.9.

Proposition 3.4. — *Let $f \in \mathcal{F}_\pi$. Then there exists a unique formal group law F_f with coefficients in A for which f is an endomorphism.*

This means

$$f(F_f(X, Y)) = F_f(f(X), f(Y)),$$

that is $f \circ F_f = F_f \circ (f \times f)$.

Proposition 3.5. — Let $f \in \mathcal{F}_\pi$ and let F_f be the corresponding group law of Proposition 3.4. Then for any $a \in A = \mathcal{O}_K$ there exists a unique $[a]_f \in A[[X]]$ such that:

- (1) $[a]_f$ commutes with f ;
- (2) $[a]_f \equiv aX \pmod{\deg 2}$.

Moreover, $[a]_f$ is then an endomorphism of the group law F_f .

From Proposition 3.5 we obtain a mapping $A \rightarrow \text{End}(F_f)$ defined by $a \mapsto [a]_f$.

For example, consider the case $K = \mathbf{Q}_p$,

$$f = pX + \binom{p}{2}X^2 + \cdots + X^p;$$

then F is the multiplicative law $X + Y + XY$, and

$$[a]_f = (1 + X)^a - 1 = \sum_{i=1}^{\infty} \binom{a}{i} X^i.$$

Proposition 3.6. — The map $a \mapsto [a]_f$ is an injective homomorphism of the ring A into the ring $\text{End}(F_f)$.

Proposition 3.7. — Let f and g be members of $\mathcal{F}_\pi$. Then the corresponding group laws are isomorphic.

3.4. Statements. — Let K be a local field and let π be a uniformizing element. Let $f \in \mathcal{F}_\pi$ and let F_f be the corresponding group law (of Proposition 3.4). We denote by $M_f = F_f(\mathbf{m}_{K_s})$ the group of points in the separable closure equipped with the group law deduced from F_f . Let $a \in A$, $x \in M_f$, and put $ax = [a]_f(x)$. By Proposition 3.6, this defines a structure of an A -module on M_f . Let E_f be the torsion submodule of M_f ; that is, the set of elements of M_f killed by a power of π .

Theorem 3.8. — The following statements hold.

- (a) The torsion submodule E_f is isomorphic (as an A -module) with K/A .
- (b) Let $K_\pi = K(E_f)$ be the field generated by E_f over K . Then K_π is an abelian extension of K .
- (c) Let u be a unit in $K^\times$. Then the element $\sigma_u = (u, K_\pi/K)$ of $G(K_\pi/K)$ acts on E_f via $[u^{-1}]_f$.
- (d) The operation described in (c) defines an isomorphism $U_K \rightarrow G(K_\pi/K)$.
- (e) The norm residue symbol $(\pi, K_\pi/K)$ is 1.
- (f) The fields K_{nr} and K_π are linearly disjoint and $K^{\text{ab}} = K_{\text{nr}} \cdot K_\pi$.

We may express the results of Theorem 3.8 as follows. We have a diagram:

$$\begin{array}{ccc}
 & K^{\text{ab}} & \\
 & \swarrow \quad \searrow & \\
 K_{\text{nr}} & & K_{\pi} \\
 & \swarrow \quad \searrow & \\
 & K &
 \end{array}$$

Here $G(K_{\text{nr}}/K) = \widehat{\mathbf{Z}}$ and $G(K_{\pi}/K) = U_K$. Moreover every $\alpha \in K^{\times}$ can be written in the form $\alpha = \pi^n u$ and σ_{α} gives σ^n (the Frobenius) on K_{nr}/K whilst σ_u gives $[u^{-1}]_f$ on K_{π}/K .

Example. Take $K = \mathbf{Q}_p$, $\pi = p$ and $f = pX + \binom{p}{2}X^2 + \cdots + X^p$. The formal group law is the multiplicative group law; E_f is the set of p^{ν} th roots of unity; K_{π} is the field denoted by $\mathbf{Q}_{p^{\infty}}$ in § 3.1, and we recover Theorems 3.1 and 3.2.

3.5. Construction of F_f , $[a]_f$. — In this section we shall construct the formal group law F_f and the map $a \mapsto [a]_f$.

Proposition 3.9. — *Let $f, g \in \mathcal{F}_{\pi}$, let n be an integer and let $\phi_1(X_1, \dots, X_n)$ be a linear form in $X_1, \dots, X_n$ with coefficients in A . Then there exists a unique $\phi \in A[[X_1, \dots, X_n]]$ such that:*

- (a) $\phi \equiv \phi_1 \pmod{\deg 2}$;
- (b) $f \circ \phi = \phi \circ (g \times \cdots \times g)$.

Remarks. (1) The property (b) may be written

$$f(\phi(X_1, \dots, X_n)) = \phi(g(X_1), \dots, g(X_n)).$$

(2) The completeness of A will not be used in the proof. Moreover, the proof shows that ϕ is the only power series with coefficients in an extension of A , which is torsion free as an A -module, satisfying (a) and (b).

Démonstration. — We shall construct ϕ by successive approximations. More precisely, we construct a sequence $(\phi^{(p)})$ such that $\phi^{(p)} \in A[[X_1, \dots, X_n]]$, $\phi^{(p)}$ satisfies (a) and (b) modulo $\deg p + 1$, and $\phi^{(p)}$ is unique modulo $\deg p + 1$.

We shall then define $\phi = \lim \phi^{(p)}$ and this will be the ϕ whose existence is asserted. We take $\phi^{(1)} = \phi_1$.

Suppose that the approximation $\phi_1 + \cdots + \phi_p = \phi^{(p)}$ has been constructed. That is,

$$f \circ \phi^{(p)} \equiv \phi^{(p)} \circ (g \times \cdots \times g) \pmod{\deg p + 1}.$$

For convenience of writing, we shall replace $g \times \cdots \times g$ by the single variable g . Now write $\phi^{(p+1)} = \phi^{(p)} + \phi_{p+1}$. Then we may write

$$f \circ \phi^{(p)} \equiv \phi^{(p)} \circ g + E_{p+1} \pmod{\deg p + 2},$$

where E_{p+1} (“the error”) satisfies $E_{p+1} \equiv 0 \pmod{\deg p + 1}$. Consider $\phi^{(p+1)}$; we have, modulo $\deg p + 2$,

$$f \circ \phi^{(p+1)} = f \circ (\phi^{(p)} + \phi_{p+1}) \equiv f \circ \phi^{(p)} + \pi \phi_{p+1}$$

(the derivative of f at the origin is π) and

$$\phi^{(p+1)} \circ g = \phi^{(p)} \circ g + \phi_{p+1} \circ g \equiv \phi^{(p)} \circ g + \pi^{p+1} \phi_{p+1} \pmod{\deg p + 2}.$$

Thus

$$f \circ \phi^{(p+1)} - \phi^{(p+1)} \circ g \equiv E_{p+1} + (\pi - \pi^{p+1}) \phi_{p+1} \pmod{\deg p + 2}.$$

These equations show that we must take

$$\phi_{p+1} = -E_{p+1}/\pi(1 - \pi^p).$$

The unicity is now clear and it remains to show that ϕ_{p+1} has coefficients in A . That is, $E_{p+1} \equiv 0 \pmod{\pi}$. Now for $\phi \in \mathbf{F}_q[[X]]$, we have $\phi(X^q) = (\phi(X))^q$ and together with $f(X) \equiv X^q \pmod{\pi}$ this gives

$$f \circ \phi^{(p)} - \phi^{(p)} \circ g \equiv (\phi^{(p)}(X))^q - \phi^{(p)}(X^q) \equiv 0 \pmod{\pi}.$$

So, given $\phi^{(p)}$, we can construct a unique $\phi^{(p+1)}$ and the proof is completed by induction and passage to the limit. $\square$

Proof of Proposition 3.4. — For each $f \in \mathcal{F}_\pi$, let $F_f(X, Y)$ be the unique solution of

$$F_f(X, Y) \equiv X + Y \pmod{\deg 2} \quad \text{and} \quad f \circ F_f = F_f \circ (f \times f),$$

whose existence is assured by Proposition 3.9. That F_f is a formal group law now requires the verification of the rules (a) to (e) above. This follows by applying Proposition 3.9: in each case one checks that the left- and the right-hand sides are solutions to a problem of the type discussed there and then uses the unicity statement of Proposition 3.9. For example, to prove associativity note that both $F_f(F_f(X, Y), Z)$ and $F_f(X, F_f(Y, Z))$ are solutions of

$$H(X, Y, Z) \equiv X + Y + Z \pmod{\deg 2}$$

and

$$H(f(X), f(Y), f(Z)) = f(H(X, Y, Z)).$$

$\square$

Proof of Proposition 3.5. — For each $a \in A$ and $f, g \in \mathcal{F}_\pi$, let $[a]_{f,g}(T)$ be the unique solution of

$$[a]_{f,g}(T) \equiv aT \pmod{\deg 2}$$

and

$$f([a]_{f,g}(T)) = [a]_{f,g}(g(T)),$$

that is $f \circ [a]_{f,g} = [a]_{f,g} \circ g$. Write $[a]_f = [a]_{f,f}$. Now we have

$$F_f([a]_{f,g}(X), [a]_{f,g}(Y)) = [a]_{f,g}(F_g(X, Y)).$$

For each side is congruent to $aX + aY$ modulo $\deg 2$, and if we replace X by $g(X)$ and Y by $g(Y)$ in either side, then the result is the same as if we substitute the sides in question in f . Thus $[a]_{f,g}$ is a formal homomorphism of F_g into F_f . If we take $g = f$, this shows that the $[a]_f$'s are endomorphisms of F_f . $\square$

Proof of Proposition 3.6. — In the same way as outlined above, one proves that

$$[a + b]_{f,g} = F_f \circ ([a]_{f,g} \times [b]_{f,g})$$

and

$$[ab]_{f,h} = [a]_{f,g} \circ [b]_{g,h}.$$

It follows from this that the composition of two homomorphisms of the type just established is reflected in the product of corresponding elements of A . Taking $f = g$, we see that the map $a \mapsto [a]_f$ is a ring homomorphism of A into $\text{End}(F_f)$. It is injective because the term of degree 1 of $[a]_f$ is aX . $\square$

Proof of Proposition 3.7. — If a is a unit in A , then $[a]_{f,g}$ is invertible (cf. the proof of Proposition 3.5) and so $F_g \simeq F_f$ by means of the isomorphism $[a]_{f,g}$. Note that $[\pi]_f = f$ and $[1]_f$ is the identity (proved as before). $\square$

This completes the proofs of Propositions 3.4, 3.5, 3.6 and 3.7.

3.6. First Properties of the Extension K_π of K . — From now on, we confine our attention to subfields of a fixed separable closure K_s of K . Given $f \in \mathcal{F}_\pi$, let F_f be the corresponding formal group law and let E_f be the torsion submodule of the A -module $F_f(\mathfrak{m}_{K_s})$. Let E_f^n be the kernel of $[\pi^n]_f$; so that $E_f = \bigcup E_f^n$. Let $K_\pi^n = K(E_f^n)$ and $K_\pi = \bigcup K_\pi^n$. If $G_{\pi,n}$ denotes the Galois group of $K(E_f^n)$ over K , then

$$G(K_\pi/K) = \varprojlim G_{\pi,n}.$$

Proposition 3.10. — (a) The A -module E_f is isomorphic to K/A ;

(b) the natural homomorphism $G(K_\pi/K) \rightarrow \text{Aut}(E_f)$ is an isomorphism.

Démonstration. — We are free to choose f as we please, since, by Proposition 3.7, different choices give isomorphic group laws. We take $f = \pi X + X^q$. Then $\alpha \in E_f^n$ if and only if $f^{(n)}(\alpha) = 0$, where $f^{(n)}$ denotes the composition $f \circ \dots \circ f$ n times; that is, $f^{(n)} = [\pi^n]_f$.

If $\alpha \in \mathfrak{m}_{K_s}$, then the equation $\pi X + X^q = \alpha$ is separable and so solvable in K_s , its solution belonging indeed to $\mathfrak{m}_{K_s}$. This shows that M_f is divisible. Hence E_f is divisible also. This already implies that E_f is a direct sum of modules isomorphic to K/A .

Let us consider the submodule E_f^1 of E_f consisting (see above) of those $\alpha \in M_f$ such that $[\pi]_f \alpha = 0$. The submodule E_f^1 is isomorphic with $A/\pi A$, since it is an A -module with q elements. This is enough to show that E_f is isomorphic to K/A .

An automorphism $\sigma \in G(K_\pi/K)$ induces an automorphism of the A -module E_f . But since $E_f \simeq K/A$ and $\text{End}_A(K/A) = A$, this gives a map

$$G(K_\pi/K) \longrightarrow \text{Aut}(E_f) = U_K.$$

This map is injective by the definition of K_π and it remains to be proved that it is surjective.

Take $n \geq 1$ and define E_f^n and K_π^n as above. We have an injection

$$G(K_\pi^n/K) \longrightarrow U_K/U_K^n,$$

where $U_K^n = 1 + \pi^n A$. Let $\alpha \in E_f^n$ be a primitive element; that is an element of E_f^n such that $[\pi^n]_f \alpha = 0$, but $[\pi^{n-1}]_f \alpha \neq 0$. Finally, we define ϕ as follows:

$$\phi = \frac{f^{(n)}}{f^{(n-1)}} = \frac{f(f^{(n-1)})}{f^{(n-1)}}.$$

Now $f = X^q + \pi X$; so $f/X = X^{q-1} + \pi$. Hence

$$\frac{f(f^{(n-1)})}{f^{(n-1)}} = (f^{(n-1)}(X))^{q-1} + \pi,$$

which is of degree $q^n - q^{n-1}$ and which is irreducible, since it is an Eisenstein polynomial. All primitive elements α are roots of ϕ . Thus the order of $G(K_\pi^n/K)$ is at least $(q-1)q^{n-1}$. On the other hand, this is actually the order of the group U_K/U_K^n . Hence $G(K_\pi^n/K) = U_K/U_K^n$. It follows that

$$G(K_\pi/K) = \varprojlim G(K_\pi^n/K) = \varprojlim U_K/U_K^n = U_K,$$

and this completes the proof of Proposition 3.10. $\square$

The same proof also yields:

Corollary 3.11. — *The element π is a norm from $K(\alpha) = K_\pi^n$.*

Démonstration. — The polynomial ϕ constructed above is monic and ends with π . Hence $N(-\alpha) = \pi$. $\square$

3.7. The Reciprocity Map. — We shall study the compositum $L = K_{\text{nr}}K_\pi$ of K_{nr} and K_π , and the symbol $(\alpha, L/K)$, $\alpha \in K^\times$. We need to compare two uniformizing elements π and $\omega = \pi u$, $u \in U_K$.

Let $\widehat{K}_{\text{nr}}$ be the completion of K_{nr} (remember: K_{nr} is an increasing union of complete fields but is not itself complete) and denote by $\widehat{A}_{\text{nr}}$ the ring of integers of $\widehat{K}_{\text{nr}}$. By definition $\widehat{K}_{\text{nr}}$ is complete; it has an algebraically closed residue field and π is a uniformizing parameter in $\widehat{K}_{\text{nr}}$. We take $f \in \mathcal{F}_\pi$ and $g \in \mathcal{F}_\omega$.

Lemma 3.12. — *Let $\sigma \in G(K_{\text{nr}}/K)$ be the Frobenius automorphism and extend it to $\widehat{K}_{\text{nr}}$ by continuity. Then there exists a power series $\phi \in \widehat{A}_{\text{nr}}[[X]]$ with $\phi(X) \equiv \varepsilon X \pmod{\deg 2}$ and ε a unit, such that*

- (a) ${}^\sigma\phi = \phi \circ [u]_f$;
- (b) $\phi \circ F_f = F_g \circ (\phi \times \phi)$;
- (c) $\phi \circ [a]_f = [a]_g \circ \phi$ for all $a \in A$.

Démonstration. — Since $\sigma - 1$ is surjective on $\widehat{A}_{\text{nr}}$ and on $\widehat{U}_{\text{nr}}$ (cf. “Corps Locaux”, p. 209), there exists a $\phi \in \widehat{A}_{\text{nr}}[[X]]$ such that $\phi(X) \equiv \varepsilon X \pmod{\deg 2}$, where ε is a unit, and ${}^\sigma\phi = \phi \circ [u]_f$. This is proved by successive approximation and we refer the reader to Lubin–Tate for the details. This particular ϕ does not necessarily give (b) and (c) but can be adjusted to do so; the computations are given in Lubin–Tate (where they appear as (17) and (18) in Lemma 2 on p. 385). Note that together the above conditions express the fact that ϕ is an A -module isomorphism of F_f into F_g . $\square$

Computation of the norm reciprocity map in L/K .

Let $L_\pi = K_{\text{nr}}K_\pi$. Since K_{nr} and K_π are linearly disjoint over K , the Galois group $G(L_\pi/K)$ is the product of the Galois groups $G(K_\pi/K)$ and $G(K_{\text{nr}}/K)$. For each uniformizing element $\pi \in A$ we define a homomorphism

$$r_\pi : K^\times \longrightarrow G(L_\pi/K)$$

such that:

- (a) $r_\pi(\pi)$ is 1 on K_π and is the Frobenius automorphism σ on K_{nr} ;
- (b) for $u \in U_K$, $r_\pi(u)$ is equal to $[u^{-1}]_f$ on K_π and is 1 on K_{nr} .

We want to prove that the field L_π and the homomorphism r_π are independent of π . Let $\omega = \pi u$ be a second uniformizing element.

First, $L_\pi = L_\omega$. For by Lemma 3.12, F_f and F_g are isomorphic over $\widehat{K}_{\text{nr}}$. Hence, the fields generated by their division points are the same. So $\widehat{K}_{\text{nr}} \cdot K_\pi = \widehat{K}_{\text{nr}} \cdot K_\omega$. On taking completions we find that the completions of $K_{\text{nr}} \cdot K_\pi$ and $K_{\text{nr}} \cdot K_\omega$ agree. In order to deduce that $K_{\text{nr}} \cdot K_\pi = K_{\text{nr}} \cdot K_\omega$ from this, we require the following:

Lemma 3.13. — *Let E be any algebraic extension (finite or infinite) of a local field and let $\widehat{E}$ be its completion. If $\alpha \in \widehat{E}$ is separable algebraic over E , then α belongs to E .*

Démonstration. — Let E_s be the separable closure of E and let E' be the adherence of E in E_s . We can view α as an element of E' . Hence it is enough to show that $E' = E$.

Let $s \in G(E_s/E)$. Since s is continuous and is the identity on E , it is also the identity on E' . Hence $G(E_s/E) = G(E_s/E')$ and by Galois theory we have $E' = E$. $\square$

It follows from Lemma 3.13 that $L_\pi = L_\omega$ and so $L_\pi = L$ is independent of π .

We turn now to the homomorphism $r_\pi : K^\times \rightarrow G(L/K)$. We shall show that $r_\pi(\omega) = r_\omega(\omega)$. This will imply that $r_\pi(\omega)$ is independent of π and so the r_π 's coincide on the local uniformizers. Since these generate $K^\times$, the result will follow.

We look first at $r_\omega(\omega)$. On K_{nr} , $r_\omega(\omega)$ is the Frobenius automorphism σ . On K_ω it is 1. On the other hand, $r_\pi(\omega)$ is σ on K_{nr} ; so we must look at $r_\pi(\omega)$ on K_ω .

Now $K_\omega = K(E_g)$, where $g \in \mathcal{F}_\omega$. Let $\phi \in \widehat{A}_{\text{nr}}[[X]]$ be as in Lemma 3.12; ϕ determines an isomorphism of E_f onto E_g . So if $\lambda \in E_g$, then we can write $\lambda = \phi(\mu)$ with $\mu \in E_f$. We look at $r_\pi(\omega)\lambda$ and we want to show that this is λ . As already remarked, $r_\pi(\omega)(\lambda) = r_\pi(\omega)\phi(\mu)$. Write $s = r_\pi(\omega)$. We want to show that $s\lambda = \lambda$; that is, ${}^s\phi(s\mu) = \phi(\mu)$.

Now $r_\pi(\omega) = r_\pi(\pi)r_\pi(u)$ and the effects of $r_\pi(\pi)$ and $r_\pi(u)$ are described in (a) and (b) above. Since ϕ has coefficients in $\widehat{K}_{\text{nr}}$,

$${}^s\phi = \sigma\phi = \phi \circ [u]_f$$

by (a) of Lemma 3.12. But

$${}^s\phi(s\mu) = \sigma\phi([u^{-1}]_f(\mu)).$$

Hence

$${}^s\phi(s\mu) = \phi \circ [u]_f \circ [u^{-1}]_f(\mu) = \phi(\mu).$$

So $r_\pi(\omega)$ is the identity on K_ω and it follows that r_π is independent of π .

Thus

$$r : K^\times \longrightarrow G(L/K)$$

is the reciprocity map θ (§ 2.7, Proposition 2.14).

All assertions of Theorem 3.8 have now been proved except the equality $L = K^{\text{ab}}$, which we are now going to prove.

3.8. The Existence Theorem. — Let K^{ab} be the maximal abelian extension of K ; it contains K_{nr} . The Existence Theorem is equivalent to the following assertion (§ 2.7, Theorem 2.11). If $I_K = G(K^{\text{ab}}/K_{\text{nr}})$ is the inertia subgroup of $G(K^{\text{ab}}/K)$, then the reciprocity map

$$\theta : U_K \longrightarrow I_K$$

is an isomorphism.

Let L be the compositum $K_\pi K_{\text{nr}}$ and let $I'_K = G(L/K_{\text{nr}})$ be the inertia subgroup of $G(L/K)$. Consider the maps

$$U_K \xrightarrow{\theta} I_K \xrightarrow{e} I'_K$$

where θ is the reciprocity map and e is the canonical map $I_K \rightarrow I'_K$. Both θ and e are surjections.

On the other hand, the composition

$$e \circ \theta : U_K \longrightarrow I'_K$$

has just been computed. If we identify I'_K with U_K it is $u \mapsto u^{-1}$. Hence the composed map $e \circ \theta$ is an isomorphism. It follows that both θ and e are isomorphisms.

As we have already noted, the first isomorphism is equivalent to the Existence Theorem. The second means that $L = K^{\text{ab}}$, since both L and K^{ab} contain K_{nr} .

Alternative Proof. Let us prove directly that every open subgroup M of $K^\times$, which is of finite index, is a norm subgroup corresponding to a finite subextension of L . This will prove both the existence theorem (§ 2.8, Theorem 2.10) and the fact that $L = K^{\text{ab}}$. Since M is open, there exists $n \geq 1$ such that $U_K^n \subset M$; since M is of finite index, there exists $m \geq 1$ such that $\pi^m \in M$; hence M contains the subgroup $V_{n,m}$ generated by U_K^n and π^m . Now let K_m be the unramified extension of K of degree m , and consider the subfield

$$L_{n,m} = K_\pi^n K_m$$

of L .

If $u \in U_K$ and $a \in \mathbf{Z}$, we know that $(u\pi^a, L_{n,m}/K)$ is equal to $[u^{-1}]$ on K_π^n and to the a th power of the Frobenius element on K_m ; hence $(u\pi^a, L_{n,m}/K)$ is trivial if and only if $u \in U_K^n$ and $a \equiv 0 \pmod{m}$, i.e. if and only if $u\pi^a \in V_{n,m}$. This shows that

$$V_{n,m} = N_{L_{n,m}/K} L_{n,m}^\times,$$

and, since M contains $V_{n,m}$, M is the norm group of a subextension of $L_{n,m}$. Q.E.D.

4. Ramification Subgroups and Conductors

4.1. Ramification Groups. — Let L/K be a galois extension of local fields with Galois group $G(L/K)$.

We recall briefly the definition of the upper numbering of the ramification groups. (For details, the reader should consult Chapter I, § 9, or “Corps Locaux”, Chap. IV.) Let the function

$$i_G : G(L/K) \longrightarrow \{\mathbf{Z} \cup \infty\}$$

be defined as follows. For $s \in G(L/K)$, let x be a generator of $\mathcal{O}_L$ as an $\mathcal{O}_K$ -algebra and put

$$i_G(s) = v_L(s(x) - x).$$

Now define G_u for all positive real numbers u by:

$$s \in G_u \quad \text{if and only if} \quad i_G(s) \geq u + 1.$$

The groups G_u are called the ramification groups of $G(L/K)$ (or of L/K). In order to deal with the quotient groups, it is necessary to introduce a second enumeration of the ramification groups called the “upper numbering”. This new numbering is given by $G^v = G_u$, where $v = \phi(u)$ and where the function ϕ is characterized by the properties:

- (a) $\phi(0) = 0$;
- (b) ϕ is continuous;
- (c) ϕ is piecewise linear;
- (d) $\phi'(u) = 1/(G_0 : G_u)$ when u is not an integer.

The G^v 's so defined are compatible with passage to the quotient: $(G/H)^v$ is the image of G^v in G/H (“Herbrand’s theorem”). This allows one to define the G^v 's even for infinite extensions.

On the other hand, we have a filtration on U_K defined by $U_K^n = 1 + \mathfrak{m}_K^n$. We extend this filtration to real exponents by $U_K^v = U_K^n$ if $n - 1 < v \leq n$.

(It should be noted that v in this context is a real number and is not to be confused with the valuation map!)

Theorem 4.1. — *Let L/K be an abelian extension with Galois group G . Then the local reciprocity map $\theta : K^\times \rightarrow G$ maps U_K^v onto G^v for all $v \geq 0$.*

Démonstration. — (1) Verification for the extensions K_π^n of § 3.6.

Let $u \in U_K^i$ with $i < n$ and $u \notin U_K^{i+1}$. Let $s = \theta(u) \in G(K_\pi^n/K)$. We have

$$i(s) = v_{K_\pi^n}(s\lambda - \lambda),$$

where λ is a uniformizing element. We choose a primitive root α for λ ; that is, an α satisfying $[\pi^n]_f \alpha = 0$ but $[\pi^{n-1}]_f \alpha \neq 0$. Observe that $s_u(\alpha) = [u^{-1}]_f \alpha$ and $u^{-1} = 1 + \pi^i v$ (see § 3.4, Theorem 3.8), where v is a unit. These imply that

$$s_u \alpha = [1 + \pi^i v]_f \alpha = F_f(\alpha, [\pi^i v]_f \alpha).$$

If we write $\beta = [\pi^i v]_f \alpha$, then β is a primitive $(n-i)$ th root (that is, $[\pi^{n-i}]_f \beta = 0$, $[\pi^{n-i-1}]_f \beta \neq 0$), and we have

$$F_f(\alpha, [\pi^i v]_f \alpha) = \alpha + \beta + \sum_{i>1, j \geq 1} \gamma_{ij} \alpha^i \beta^j$$

for some $\gamma_{ij} \in \mathcal{O}_K$. Accordingly,

$$s_u(\alpha) - \alpha = \beta + \sum \gamma_{ij} \alpha^i \beta^j$$

and

$$v_{K_\pi^n}(s_u(\alpha) - \alpha) = v_{K_\pi^n}(\beta).$$

Now α is a uniformizing element in K_π^n , whilst β is a uniformizing element in K_π^{n-i} and K_π^n/K_π^{n-i} is totally ramified. Its degree is q^i . So we have determined the i function of $\theta(u)$; namely, if $u \in U_K^i$ but $u \notin U_K^{i+1}$, then

$$i(\theta(u)) = q^i.$$

This says that if $q^{i-1} - 1 < u \leq q^i - 1$, then the ramification group G_u is $\theta(U_K^i)$.

We turn now to the upper numbering of the G_u 's. That is, we define a function

$$\phi = \phi_{K_\pi^n/K}$$

corresponding to the extension K_π^n/K , which satisfies the conditions (a) to (d) above. Namely,

$$\phi(u) = \int_0^u \frac{dt}{(G_0 : G_t)}.$$

The graph of $\phi(u)$ is shown in Figure 1.

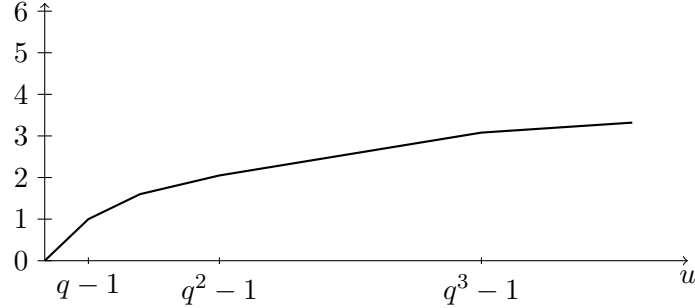


FIGURE 1.

Then $G^v = G_u$ with $v = \phi(u)$. If $q^{i-1} - 1 < u \leq q^i - 1$, then

$$\phi'(u) = \frac{1}{q^i - q^{i-1}} \quad \text{and} \quad (U_K : U_K^i) = q^i - q^{i-1}.$$

So if $i - 1 < v \leq i$, then $G^v = \theta(U_K^v)$ for $v \leq n$.

(2) Verification in the general case.

Having proved the theorem for K_π^n , it follows for $K_\pi = \bigcup K_\pi^n$ by taking projective limits. Hence also for $K_\pi K_{\text{nr}}$, since both extensions have the same inertia subgroup. Since $K_\pi K_{\text{nr}}$ is the maximal abelian extension, the result is true in general.

This concludes the proof of the theorem. $\square$

Corollary 4.2. — *The jumps in the filtration $\{G^v\}$ of G occur only for integral values of v .*

Démonstration. — This follows from Theorem 4.1, since it is trivial for filtrations of U_K and the theorem transforms one into the other.

[This result is in fact true for any field which is complete with respect to a discrete valuation and which has perfect residue field (theorem of Hasse–Arf), cf. “Corps Locaux”, Chap. IV, V.] $\square$

4.2. Abelian Conductors. — Let L/K be a finite abelian extension and let $\theta : K^\times \rightarrow G(L/K)$ be the corresponding reciprocity map. There is a smallest number n such that $\theta(U_K^n) = 0$. This number n is called the conductor of the extension L/K and is denoted by $f(L/K)$.

Proposition 4.3. — *Let c be the largest integer such that the ramification group G_c is not trivial. Then*

$$f(L/K) = \phi_{L/K}(c) + 1.$$

Démonstration. — This is a trivial consequence of Theorem 4.1 and the fact that the upper numbering is obtained by applying ϕ . $\square$

Now let L/K be an arbitrary galois extension. Let $\chi : G \rightarrow \mathbf{C}^\times$ be a one-dimensional character and let L_χ be the subfield of L corresponding to $\text{Ker}(\chi)$. The field L_χ is a cyclic extension of K and $f(L_\chi/K)$ is called the conductor of χ and is denoted by $f(\chi)$.

Proposition 4.4. — *Let $\{G_i\}$ be the ramification subgroups of $G = G(L/K)$ and write $g_i = \text{Card}(G_i)$. Then*

$$f(\chi) = \sum_{i=0}^{\infty} \frac{g_i}{g_0} (1 - \chi(G_i)),$$

where

$$\chi(G_i) = g_i^{-1} \sum_{s \in G_i} \chi(s)$$

is the “mean value” of χ on G_i .

Démonstration. — We have $\chi(G_i) = 1$ if χ is trivial on G_i (that is, equal to 1 everywhere) and $\chi(G_i) = 0$ if χ is non-trivial on G_i . Hence (the reader is referred to “Corps Locaux”, Chaps. IV and VI for the details)

$$\sum_{i=0}^{\infty} \frac{g_i}{g_0} (1 - \chi(G_i)) = \sum_{i=0}^{c_\chi} \frac{g_i}{g_0} = \phi_{L/K}(c_\chi) + 1,$$

where c_χ is the largest number such that the restriction $\chi|_{G_{c_\chi}} \neq 1$. Now $f(\chi) = f(L_\chi/K)$ is equal to $\phi_{L_\chi/K}(c) + 1$, where c is defined as in Proposition 4.3 for the extension L_χ/K .

Since $\phi_{L/K}$ is transitive, it suffices to show that $c = \phi_{L/L_\chi}(c_\chi)$, and this is a consequence of Herbrand’s theorem (§ 4.1). $\square$

4.3. Artin’s Conductors. — Let L/K be a finite galois extension with Galois group $G = G(L/K)$. Let χ be a character of G (that is, an integral combination of irreducible characters). Artin defined the conductor of χ as the number

$$f(\chi) = \sum_{i=0}^{\infty} \frac{g_i}{g_0} (\chi(1) - \chi(G_i)).$$

If χ is irreducible of degree 1, this $f(\chi)$ coincides with the previous $f(\chi)$.

We define Artin’s character a_G as follows. For $s \in G$, set

$$a_G(s) = -f i_G(s) \quad \text{if } s \neq 1, \quad a_G(1) = f \sum_{s \neq 1} i_G(s).$$

Here f is the residue degree $[l : k]$ (not to be confused with the conductor!), and i_G is the function defined above.

Proposition 4.5. — *Let $g = \text{Card}(G)$. Then*

$$f(\chi) = (a_G, \chi) = \frac{1}{g} \sum_{s \in G} \chi(s) a_G(s).$$

Démonstration. — For $s \neq 1$, the integer $i_G(s)$ is the number of indices $i \geq 0$ for which $s \in G_i$. Hence

$$\sum_{s \neq 1} i_G(s) = \sum_{i \geq 0} (g_i - 1)$$

and

$$\sum_{s \neq 1} \chi(s) i_G(s) = \sum_{i \geq 0} \left(\sum_{s \in G_i} \chi(s) - \chi(1) \right).$$

Using the definition of a_G , we obtain

$$\begin{aligned} \frac{1}{g} \sum_{s \in G} \chi(s) a_G(s) &= \frac{f}{g} \sum_{i \geq 0} \left(\chi(1)(g_i - 1) - \sum_{s \in G_i} \chi(s) + \chi(1) \right) \\ &= \frac{f}{g} \sum_{i \geq 0} \left(g_i \chi(1) - \sum_{s \in G_i} \chi(s) \right). \end{aligned}$$

Since $g = f g_0$, this is

$$\sum_{i \geq 0} \frac{g_i}{g_0} \left(\chi(1) - g_i^{-1} \sum_{s \in G_i} \chi(s) \right) = \sum_{i \geq 0} \frac{g_i}{g_0} (\chi(1) - \chi(G_i)).$$

By definition the last expression is $f(\chi)$. This proves the formula. $\square$

Proposition 4.6. — (a) *Let $K \subset L' \subset L$ be a tower of galois extensions, let χ' be a character of $G(L'/K)$ and let χ be the corresponding character of $G(L/K)$. Then $f(\chi) = f(\chi')$.*

(b) *Let $K \subset K' \subset L$ and let ψ be a character of $G(L/K')$ and let ψ^* be the corresponding induced character of $G(L/K)$. Then*

$$f(\psi^*) = \psi(1) v_K(\mathfrak{d}_{K'/K}) + f_{K'/K} f(\psi),$$

where $f_{K'/K}$ is the residue degree of K'/K and $\mathfrak{d}_{K'/K}$ is the discriminant of K'/K .

Démonstration. — The proof depends on properties of the i_G function and on the relation between the different and the discriminant, and can be found in “Corps Locaux”, Chap. VI. $\square$

Theorem 4.7 (Artin). — *Let χ be the character of a representation of G . Then $f(\chi)$ is a positive integer.*

Démonstration. — Let χ be the character of the rational representation M of G . It follows from representation theory that

$$\chi(1) = \dim M$$

and

$$\chi(G_i) = \dim M^{G_i}.$$

Thus in

$$\sum \frac{g_i}{g_0} (\chi(1) - \chi(G_i)),$$

each term is positive (≥ 0) and so $f(\chi) \geq 0$.

It remains to be proved that $f(\chi)$ is an integer. According to a theorem of Brauer, χ can be written

$$\chi = \sum m_i \psi_i^*,$$

where $m_i \in \mathbf{Z}$ and ψ_i^* is induced by a character ψ_i of degree 1 of a subgroup H_i of G .

Hence, since

$$f(\psi_i^*) = \psi_i(1) v_K(\mathfrak{d}_{K_i/K}) + f_{K_i/K} f(\psi_i),$$

$f(\psi_i^*)$ is an integer provided that $f(\psi_i)$ is. But since ψ_i has degree 1, $f(\psi_i)$ may be interpreted as an abelian conductor and so is obviously an integer. This proves the theorem. $\square$

4.4. Global Conductors. — Let L/K be a finite galois extension of number fields and let $G = G(L/K)$ be the Galois group. If χ is a character of G , then we define an ideal $\mathfrak{f}(\chi)$ of K , the conductor of χ , as follows. Let $\mathfrak{p}$ be a prime ideal in K and choose a prime ideal $\mathfrak{P}$ in L which divides $\mathfrak{p}$. Let

$$G_{\mathfrak{p}} = G(L_{\mathfrak{P}}/K_{\mathfrak{p}})$$

be the corresponding decomposition subgroup. Let $f(\chi, \mathfrak{p})$ be the Artin conductor of the restriction of χ to $G_{\mathfrak{p}}$, as defined above. We have $f(\chi, \mathfrak{p}) = 0$ when $\mathfrak{p}$ is unramified. The ideal

$$\mathfrak{f}(\chi) = \prod_{\mathfrak{p}} \mathfrak{p}^{f(\chi, \mathfrak{p})}$$

is called the (global) conductor of χ .

In this notation, Proposition 4.6 gives:

Proposition 4.8. — *Let K'/K be a sub-extension of L/K . Let ψ be a character of $H = G(L/K')$ and let ψ^* be the induced character of $G(L/K)$. Then*

$$\mathfrak{f}(\psi^*) = \mathfrak{d}_{K'/K}^{\psi(1)} N_{K'/K}(\mathfrak{f}(\psi)),$$

where $\mathfrak{d}_{K'/K}$ is the discriminant of K'/K .

We apply Proposition 4.8 to the case $\psi = 1$ and we denote the induced character ψ^* by $s_{G/H}$ (it corresponds to the permutation representation of G/H). Since $\mathfrak{f}(\psi) = (1)$ we obtain:

Corollary 4.9. — *We have*

$$\mathfrak{f}(s_{G/H}, L/K) = \mathfrak{d}_{K'/K}.$$

In the case $H = 1$ we have $s_{G/H} = r_G$, the character of the regular representation of G , and the corollary reads

$$\mathfrak{d}_{L/K} = \prod_{\chi} \mathfrak{f}(\chi)^{\chi(1)},$$

where χ runs through the set of irreducible characters of G . This is the “Führerdiskriminantenproduktformel” of Artin and Hasse, which was first proved by analytical methods (L functions). In the abelian case it reads:

$$\mathfrak{d}_{L/K} = \prod_{\chi: G \rightarrow \mathbf{C}^\times} \mathfrak{f}(\chi).$$

In the quadratic case it reduces to the fact that the discriminant is equal to the conductor.

4.5. Artin’s Representation. — We return to the local case.

Theorem 4.10. — *Let L/K be a finite galois extension of local fields with Galois group G . Let a_G be the Artin character of G defined above (cf. § 4.3). Then a_G is the character of a complex linear representation of G called “the” Artin representation.*

Démonstration. — The character a_G takes the same values on conjugate elements and so is a class function. It follows that a_G is a combination

$$\sum_{\chi} m_{\chi} \chi,$$

with complex coefficients m_{χ} , of the irreducible characters χ . Since

$$m_{\chi} = (a_G, \chi) = f(\chi),$$

we know (Proposition 4.5 and Theorem 4.7) that m_χ is a positive integer. Hence the result.

Now let V_χ be an irreducible representation corresponding to χ . We can define Artin's representation A_G by:

$$A_G = \sum f(\chi) \cdot V_\chi,$$

where the summation is over all irreducible characters χ . □

Remark 4.11. — This construction of A_G is rather artificial. Weil has posed the problem of finding a “natural” A_G .

Theorem 4.12. — *Let ℓ be a prime number not equal to the residue characteristic. Then the Artin representation can be realized over $\mathbf{Q}_\ell$.*

Démonstration. — See J.-P. Serre, *Annals of Mathematics*, 72 (1960), 406–420, or “Introduction à la théorie de Brauer”, Séminaire I.H.E.S., 1965/66. □

Examples exist where the Artin representation cannot be realized over $\mathbf{Q}$, $\mathbf{R}$ or $\mathbf{Q}_p$, where p is the residue characteristic. This suggests that there is no trivial definition of the Artin representation.

Assume now that L/K is totally ramified. Let $u_G = i_G - 1$; we have

$$u_G(s) = -1 \quad \text{if } s \neq 1, \quad u_G(s) = \text{Card}(G) - 1 \quad \text{if } s = 1.$$

Now

$$a_G = u_G + b_G$$

where b_G is a character of some representation.

Note. $a_G = u_G$ if and only if L/K is tamely ramified. So b_G is a measure of how wild the ramification is.

Theorem 4.13. — *Let ℓ be a prime number not equal to the residue characteristic. Then there exists a finitely generated, projective $\mathbf{Z}_\ell[G]$ module $B_{G,\ell}$ with character b_G and this module is unique up to isomorphism.*

Démonstration. — This follows from a theorem of Swan, “Topology”, 2 (1963), Theorem 5, combined with Theorem 4.12 above and the remark that $b_G(s) = 0$ when the order of s is divisible by ℓ . [See also the I.H.E.S. seminar quoted above.] □

For applications of Theorem 4.13 to the construction of invariants of finite G -modules, see M. Raynaud, “Sém. Bourbaki”, 1964/65, exposé 286.

These invariants play an important role in the functional equation of the zeta functions of curves.

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